

1 PRODUCT INTRODUCTION

1: Product Appearance



2: Technical Specifications

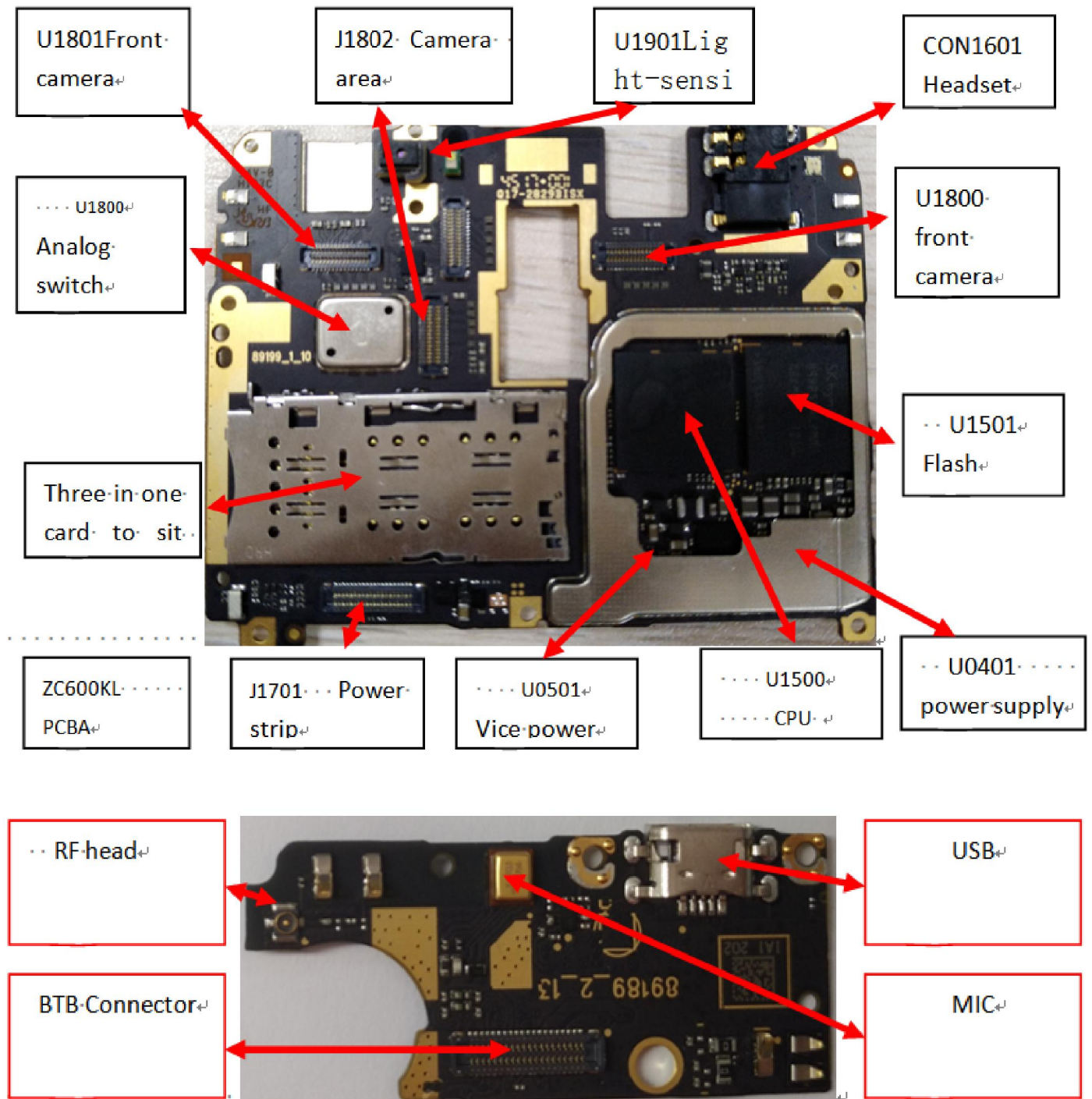
Zenfone selfie plusZC600KL_T89189 Product Spec	
ID	D style , Double side glass (2.5D), Plastic middle frame
Dimensions	Dimensions ≤ 160.62*76.15 *7.80.73mm
Color	for Qualcomm Snapdragon CPU430 MSM8937-9 435 MSM8940:5A , 5B , 5C for CPU630: 5A , 5B ,5C
Weight	168.3g
OS	WW: Android N, Zen UI full set CN: Andorid N
Processor	for Qualcomm Snapdragon CPU430 MSM8937-9, pls refer to CPU430 RF band sheet for Qualcomm CPU630 SDM 630 pls find "MKT-Spec RF Band with CPU 630 " sheet
Modem	GSM/GPRS/EDGE; WCDMA/HSPA+/DC-HSPA+; FDD-LTE, DSDS; CPU430 cat 4/CPU630 Cat6
Peak Data Rate	DC-HSPA+ (DL/UL): 42/5.76 Mbps; LTE DL CAT13 (400Mbps)/ UL CAT5 (75Mbps)
VoLTE	Support;VoLTE
RF Band Support (TBD)	for Qualcomm Snapdragon CPU430 MSM8937-9, pls refer to CPU430 RF band sheet for Qualcomm CPU630 SDM 630 pls find "MKT-Spec RF Band with CPU 630 " sheet
RAM/Memory	EMCP 3/32 & 4/64 SDM630@LPDDR4 1333MHZ MSM8937@LPDDR3 800MHZ
Micro SD Socket	Micro SD Card Support (Up to 2TB) (SD Card slot * 1)
Display	5.99" TFT LED Backlight (LCM : 400 nits (MIN); CG+LCM : 344nits (MIN)); Incell,18:9 IPS FHD2160*1080 ,colay HD (切角) 720*1440(SR後pending) Cover Lens, 2.5D glass Capacitive touch panel with 10 points multi-touch (support Glove touch) , AS Coating touch performance minimun requirement: 60ms response time
Front Camera	F2.0 20M FF IMX376-4C+ F2.4 8M 120
Main Camera	F2.2 16M AF OV16880/OV16885 +8M F2.4 120
Battery	Li-Polymer,Non-removable; TYPE: 3300mAh ;MIN: 3150mAh; Nominal Voltage: 3.85V; Charge Voltage : 4.40V;
Sensor	1. Accelerator 2. E-Compass 3. Proximity 4. Hall sensor 5. Ambient light sensor 6. Gyroscope
Fingerprint	Support ; Back ; Coating Save up to 5 fingerprints/ 0.3sec unlock/ 360 degree sensing/ Wet finger identification CN SKU: Support Fingerprint Payment (Alipay, Soter)

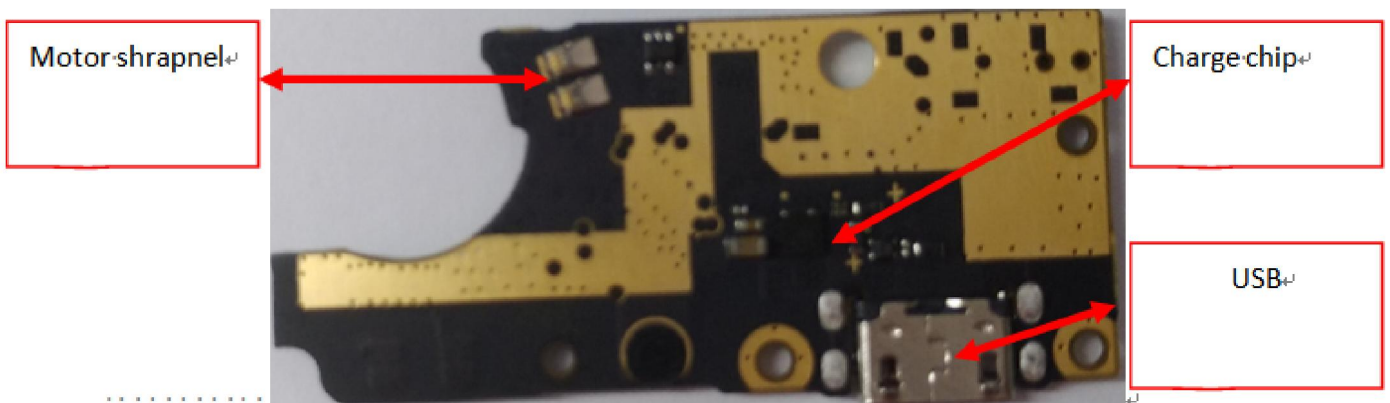
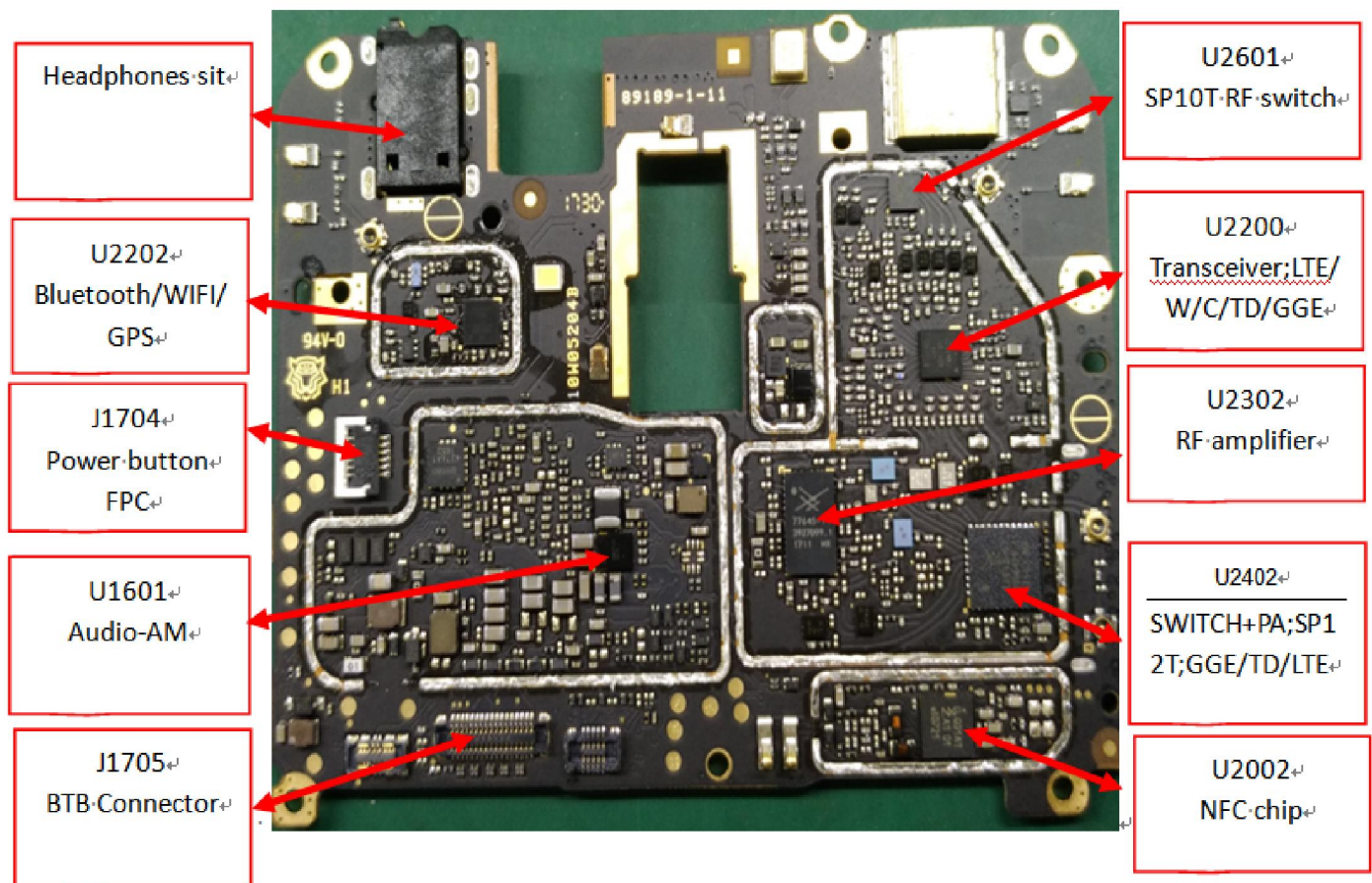
WLAN	Integrated WiFi-Direct CPU 430 WLAN 802.11 b/g/n; CPU630 802.11 a/b/g/n/ac		
VoWiFi	Support		
Bluetooth	Bluetooth V 4.1 (EDR + A2DP)		
	Bluetooth Profile: A2DP + AVRCP + HID + PAN + OPP		
NFC	YES.		
	1. Mobile payment		
	2. Reader mode		
	3. P2P mode		
FM Radio	Support		
GPS	Support GPS, AGPS, Glonass, BDS		
Interface	Mirco USB2.0 (support OTG)		
	Audio jack: Head phone		
OTG	Support OTG function but W/T bundleOTG cable		
Flash LED	Front Camera: Single LED Flash		
	Rear Camera: Single LED Flash		
Vibrator	Yes .Type: 1027		
Audio	PMIC integrated Audio CODEC		
	大磁Speaker; Type: 1115		
Microphone	Microphone*2(one for de-noise)		
SIM	DSDS, Nano SIM+Nano SIM + SD Card*1 (DSDS 4G+3G (Sim1, Sim2 support LTE), 3 in 3		
Button	Power and Volume Up/Down		
Charger	5V/1A for Qualcomm Snapdragon CPU430 5V/2A for Qualcomm CPU630 Support JEITA		
Reverse Charging	No		
Detection Requirements		SD430	SDM630
	1.PCB ID Recognize	NO	NO
	2. USBLow impedance detection	NO	NO
	3. USBHigh temperature alarm	YES	YES
	4. Charger type identification	NO	YES
	5. hardwareJeita	YES	YES
	6. OTG Recognize	YES	YES
	7. Battery ID identification	YES	YES
	8.SAR	YES	YES
Accessories in Gift Box	Giftbox, User Manual, Warranty Card, Micro USB 2.0 cable, Adaptor, Earphone etc. Packing , Ejector Pin etc.		
Accessory	clear soft bumper		
Waterproof	YES, IPX2		
Remark	all follow ASUS standard, CPU630: 16M & 20M support 4K video(3840*2160, maximum resolution of up to 24 MP at 30 fps)		

3: CAUTIONS

- I. Flashing & Servicing must be undertaken by qualified personnel only.
- II. Ensure all work is carried out at an anti-static workstation and that an anti-static wrist strap is worn.
- III. Use only approved Tools & components as specified in the parts list.
- IV. Ensure all components, modules, screws, and insulators are correctly re-fitted after servicing and alignment
- V. Ensure all cables and wires are repositioned correctly if Handset disassembled
- VI. Electrostatic discharge can easily damage the sensitive components of electronic products. Therefore, Service Centre must adhere the precautions which mentioned above.

4: service Details





2 Maintenance Tools

	Name: Constant-temperature Hot Air Gun Purpose: Heat component.
	Name: Constant-temperature Hot Air Gun Purpose: Heat component.
	Name: Soldering Iron Purpose : Repair soldering
	Name: DC Power Supply Purpose: Supply voltage and current.
	Name: Soldering Station Purpose: Fix main board.
	Name: Pb-free Solder Wire Purpose: Join component.
	Name: Digital Multimeter Purpose: Service measure
	Name: Oscilloscope Purpose: Service measure



Name: Tool kit

Purpose : Assemble and Disassemble tool

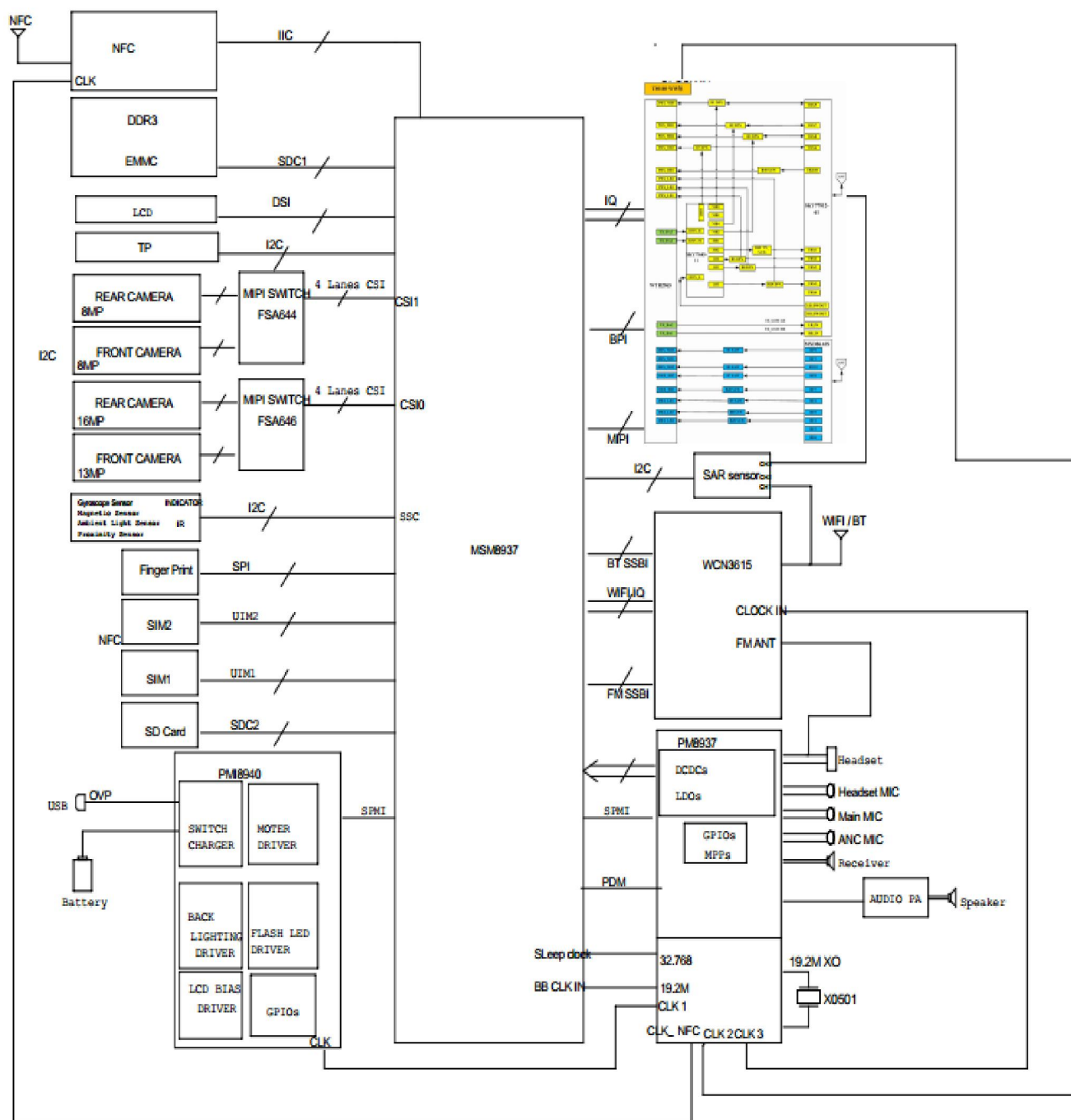


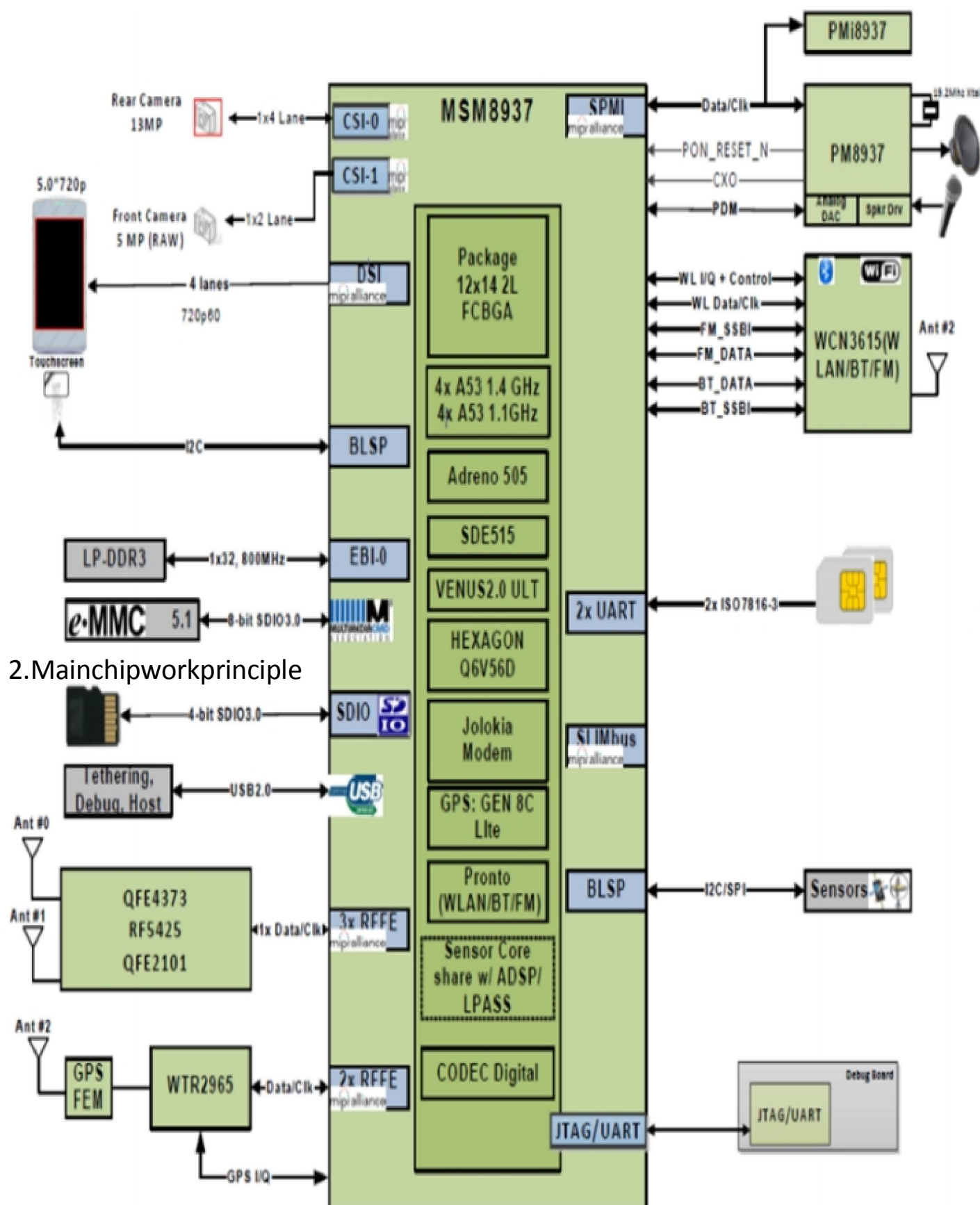
Name: Electronic Screw Driver

Purpose : Assemble and Disassemble tool

Internal block diagram

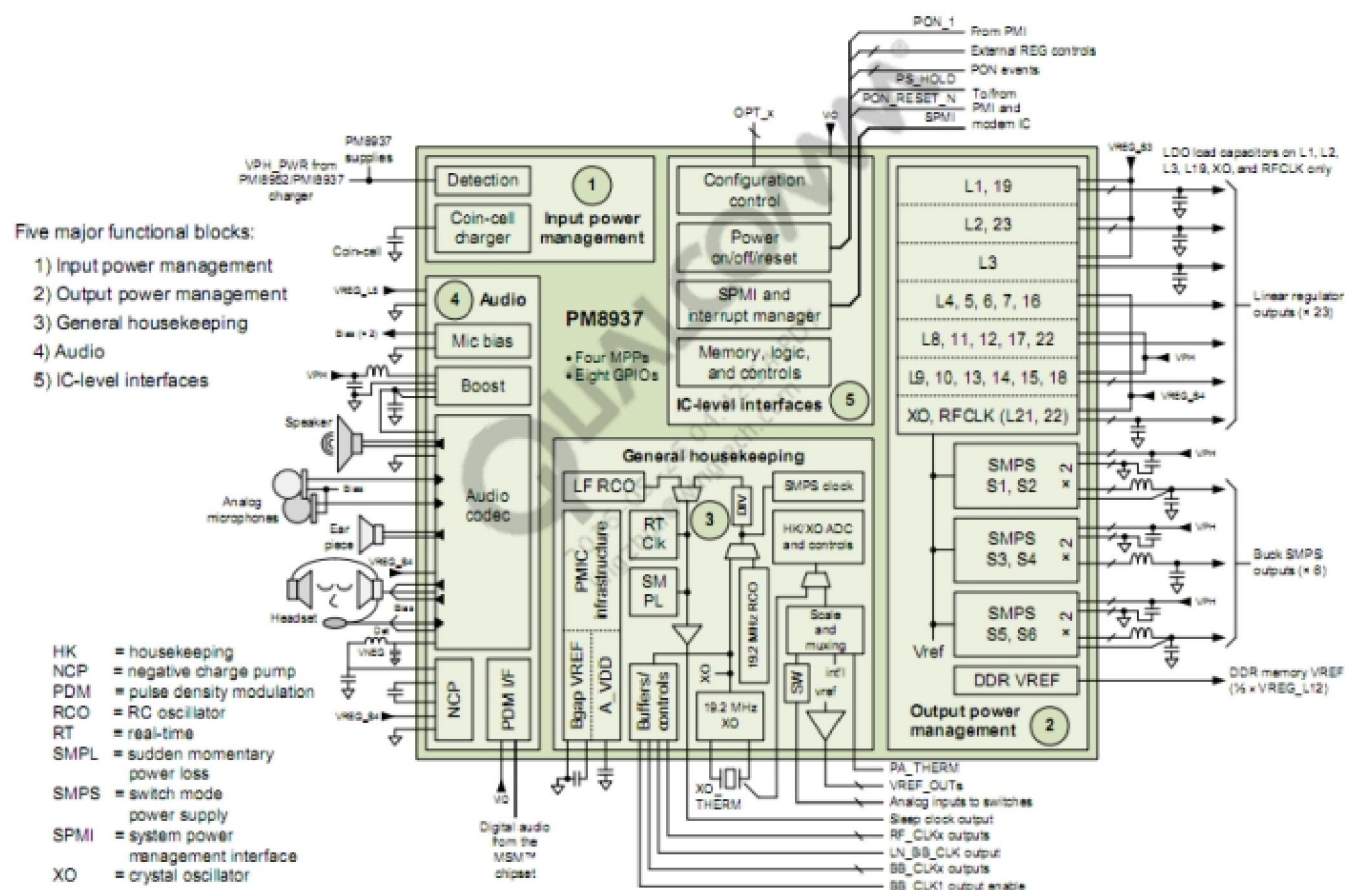
System Module Block Diagram



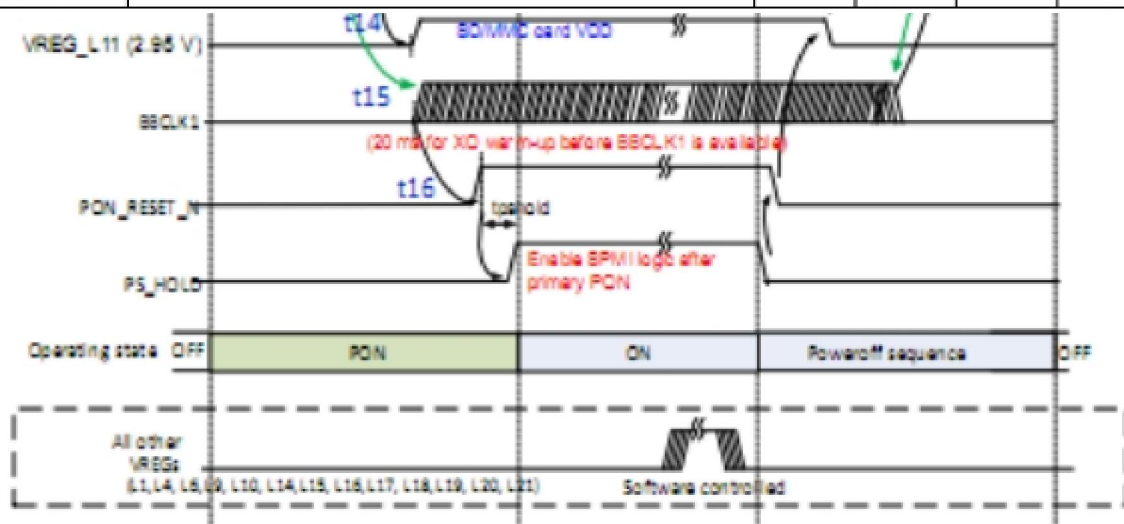


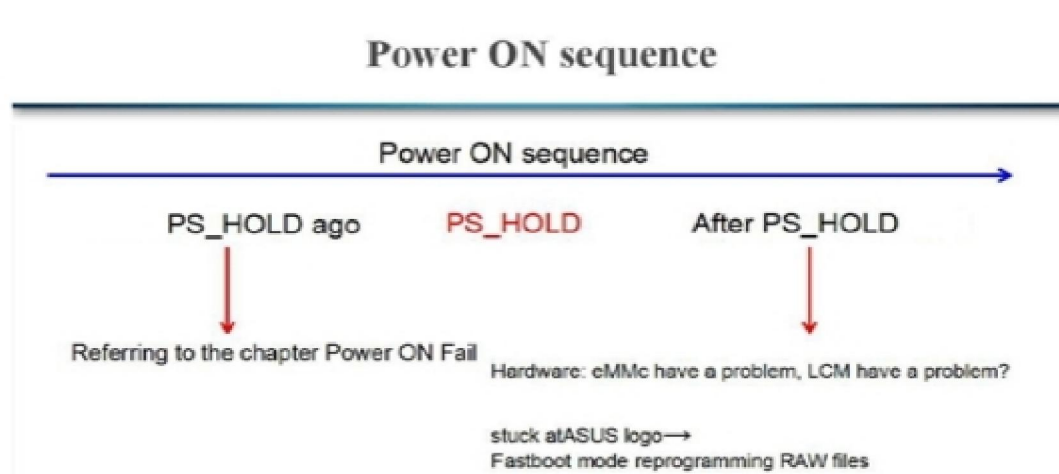
2. Main chip work principle

Function	Circuit type	Default V (V) ¹	Specified range (V) ² (MSM8937)	Programmable range (V)	Rated current (mA)	Default on	Expected use (MSM8937)
S1	ULT-SMPS	1.225	0.900–1.350	0.375–1.5625	2000	N	MSM modem
S2	ULT-SMPS	1.225	0.550–1.350	0.375–1.5625	3000	Y	MSM core and graphics
S3	HF-SMPS	1.288	1.200–1.4125	0.375–1.5625	2700	Y	Low-voltage LDOs (1, 2, 3, 19, and 23)
S4	ULT-SMPS	2.050	1.800–2.050	1.550–3.126	2500	Y	High-voltage LDOs (4, 5, 6, 7, 16, RFCLK, and XO)
S5	FT-SMPS	1.225	1.050–1.350	0.350–1.355	3000	Y	MSM applications processor
S6	FT-SMPS	1.225	1.050–1.350	0.350–1.355	3000	N	MSM applications processor
L1	NMOS LDO	1.000	1.000	0.375–1.5375	1200	N	RFICs
L2	NMOS LDO	1.200	1.200	0.375–1.5375	1200	Y	LPDDR2/LPDDR3, MIPI CSI, and DSI
L3	NMOS LDO	1.225	0.750–1.350	0.375–1.5375	1200	Y	VDDMX
L4	PMOS LDO	1.800	1.800	1.750–3.3375	450	N	RFICs and GPS eLNA
L5 ³	PMOS LDO	1.800	1.800	1.750–3.3375	500	Y	Most digital I/Os, MSM pad groups 3 and 7, LPDDR, and eMMC
L6	PMOS LDO	1.800	1.800	1.750–3.3375	300	N	MSM DSI PLL and OTP, camera, touchscreen, display, and sensors
L7	PMOS LDO	1.800	1.800	1.750–3.3375	150	Y	MSM analog and PLLs, WCN XO, and PM baseband clock driver
L8	PMOS LDO	2.900	2.900	1.750–3.3375	600	Y	eMMC
L9	PMOS LDO	$V_{out} = 3.3 \text{ V}$ for $V_{BAT} > 3.575 \text{ V}$; $V_{out} = 3 \text{ V}$ for $V_{BAT} < 3.575 \text{ V}$	3.000–3.300	1.750–3.3375	600	N	WCN
L10	PMOS LDO	2.8	2.800	1.750–3.3375	150	N	Sensors
L11 ⁴	PMOS LDO	2.950	2.950	1.750–3.3375	800	Y	Micro SD
L12 ³	PMOS LDO	2.950	1.800/2.950	1.750–3.3375	150	Y	MSM pad group 2 and SDC2
L13	PMOS LDO	3.075	3.075	1.750–3.3375	50	Y	MSM USB and audio



Parameter	Comments	Min	Typ ²	Max	Units
t(settle) ³	Regulator settling time	–	–	300	μs
t(xo)	XO warm-up time	–	15	18	ms
t(ps_hold) ^{4, 5} ■ Default timeout	PS_HOLD timeout	–	–	240	ms
t(off)	Time between regulator disable signals	–	256	600	μs
KPDPWR_N initiated poweron					
KPDPWR_N	–	0	0	0	ms
VREG_S4 (t1)	–	–	45	–	ms
VREG_S3 (t2)	–	–	1.8	–	ms
GPIO4 (t3)	–	–	0.5	–	ms
VREG_L3 (t4)	–	–	0.3	–	ms
VREG_S2 (t5)	–	–	0.3	–	ms
VREG_S5 (t6)	–	–	0.55	–	ms
MPP1 (t7)	–	–	0.33	–	ms
VREG_L5, BBLOCK1_EN (t8)	–	–	0.33	–	ms
VREG_XO	–	–	–	–	ms
VREG_L7 (t9)	–	–	0.1	–	ms
VREG_L2 (t10)	–	–	0.2	–	ms
VREG_L13 (t11)	–	–	0.83	–	ms
VREG_L8 (t12)	–	–	0.22	–	ms
VREG_L12 (t13)	–	–	0.22	–	ms
VREG_L11 (t14)	–	–	0.22	–	ms
BBLOCK (t15)	–	–	16	–	ms
PON_RESET_N (t16)	–	–	0.275	–	ms





1 boot process instructions

1. When the battery is connected to the battery connector, VBAT is generated and provided to the PMU (U0100). The PMU

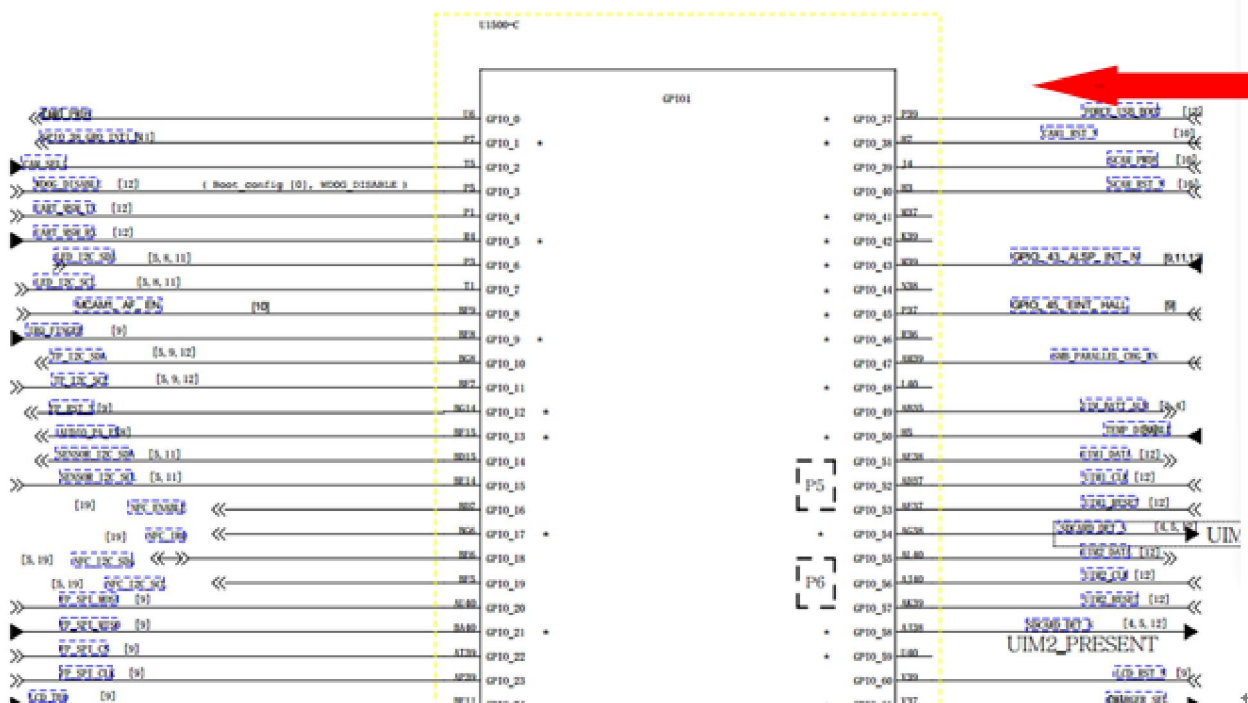
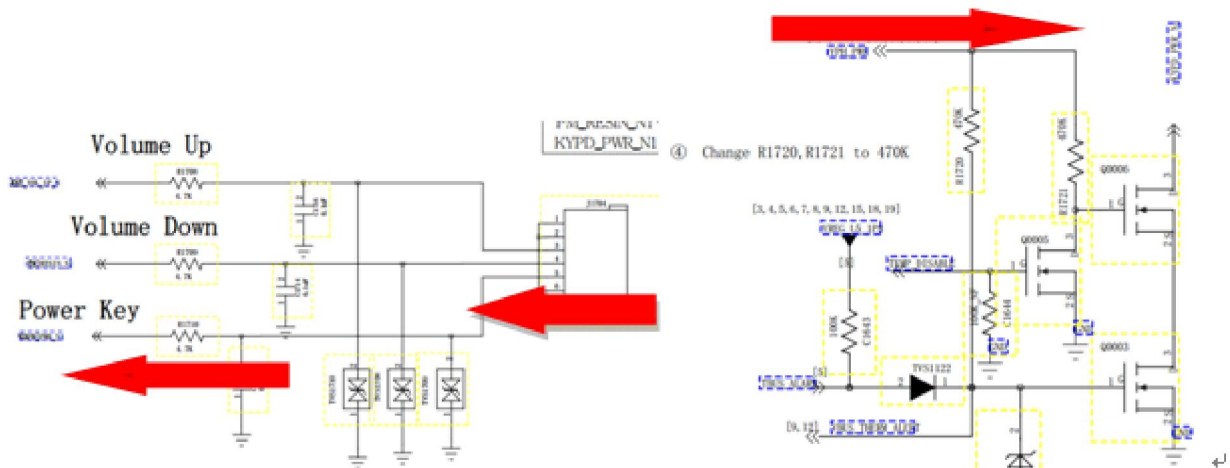
provides a power-on detection voltage of 'PWRKEY'. The 32.768KHz crystal begins to vibrate.

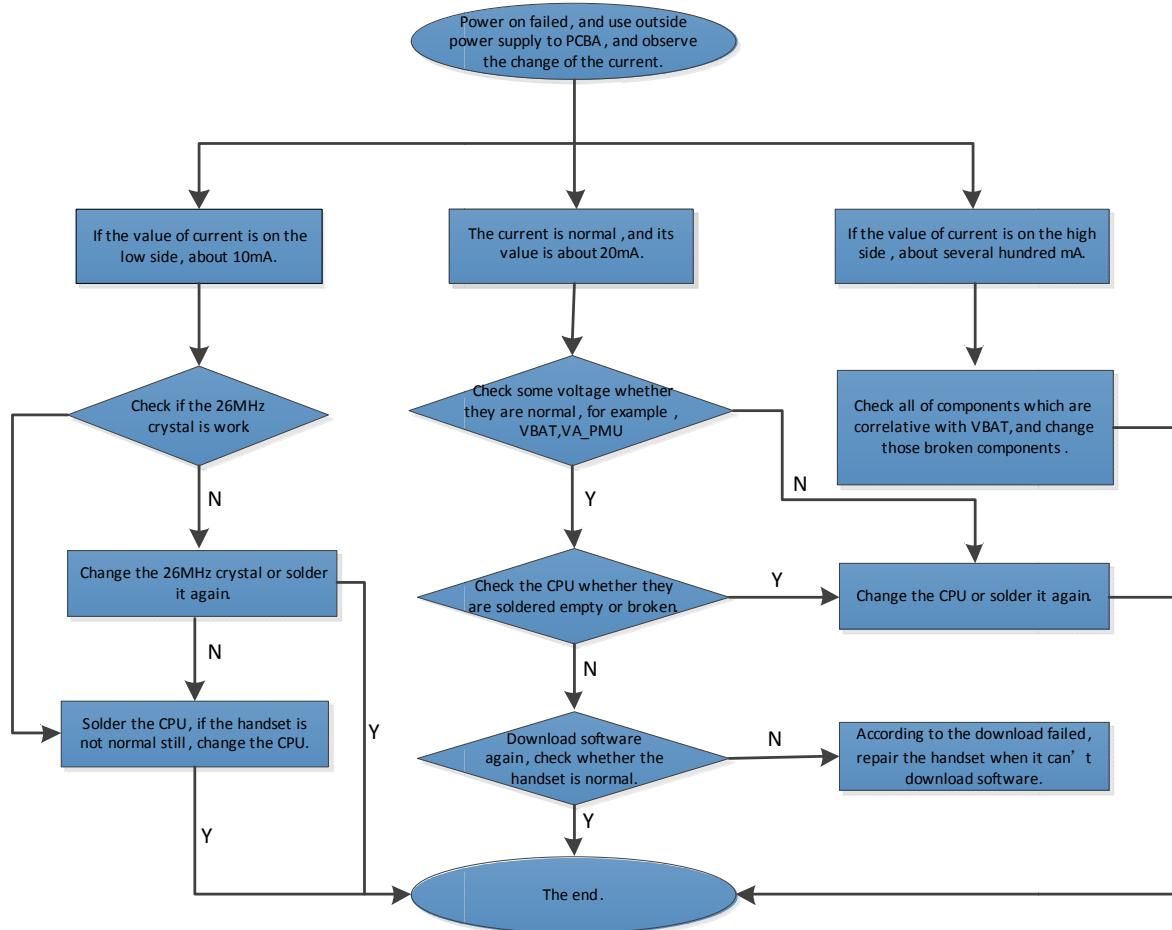
2. Long press the power button will be 'PWRKEY' pulled low to achieve the boot detection, then start the output voltage of each PMU. At this point, the baseband is ready to boot, the CPU sends BBWAKEUP signal back to the PMU. Because the boot detection has a certain delay to prevent misuse, press the power button to maintain a certain time, and, when the voltage output, the system reset to ensure the correct function and timing.

3. The 'VTCXO_1_PMU' voltage is supplied to oscillator X0200 via 'LTE_VTCXO1'. The 32.768KHz oscillator is started up and the clock is passed to the Transceiver and the system. After several clock cycles, the PMU output resets the CPU command.

4. After the CPU is initialized, read the power-on command from the ROM or send a signal through the serial port to communicate with the download computer. After the computer information feedback, handshake is achieved and the software download starts to eMMC.

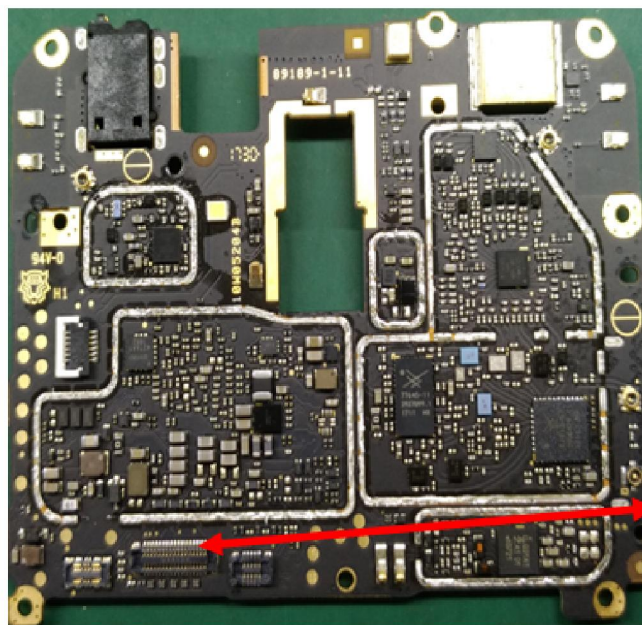
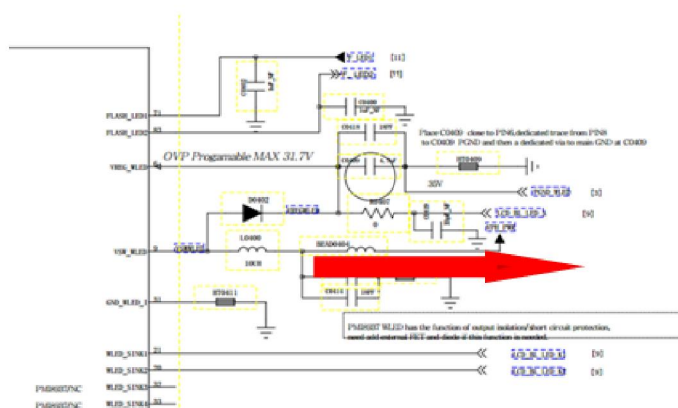
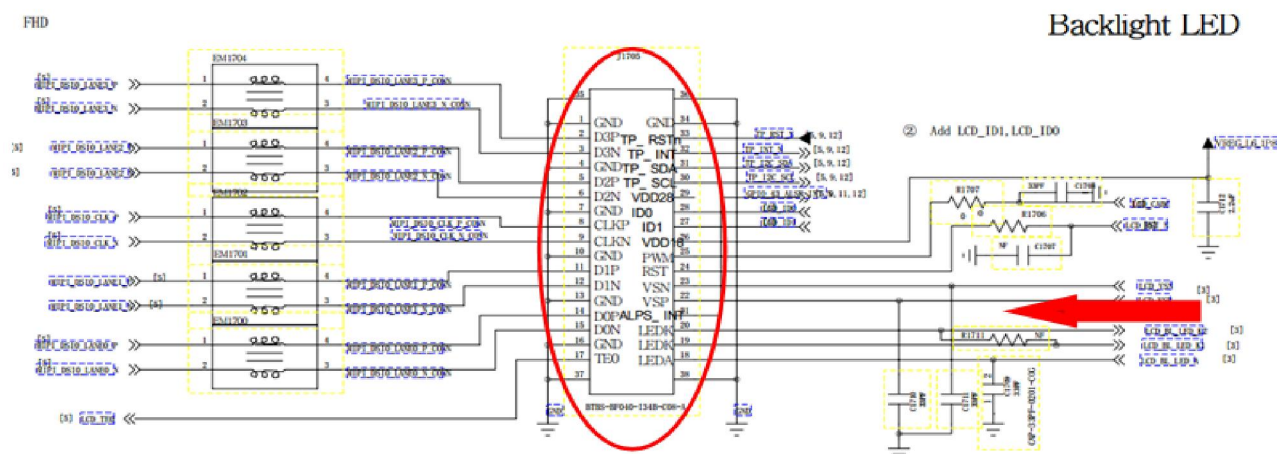
5. If you do not connect the download line, the phone runs the boot process.





2.Display issue/No display

Block Diagram



J1705
BTB-Connector

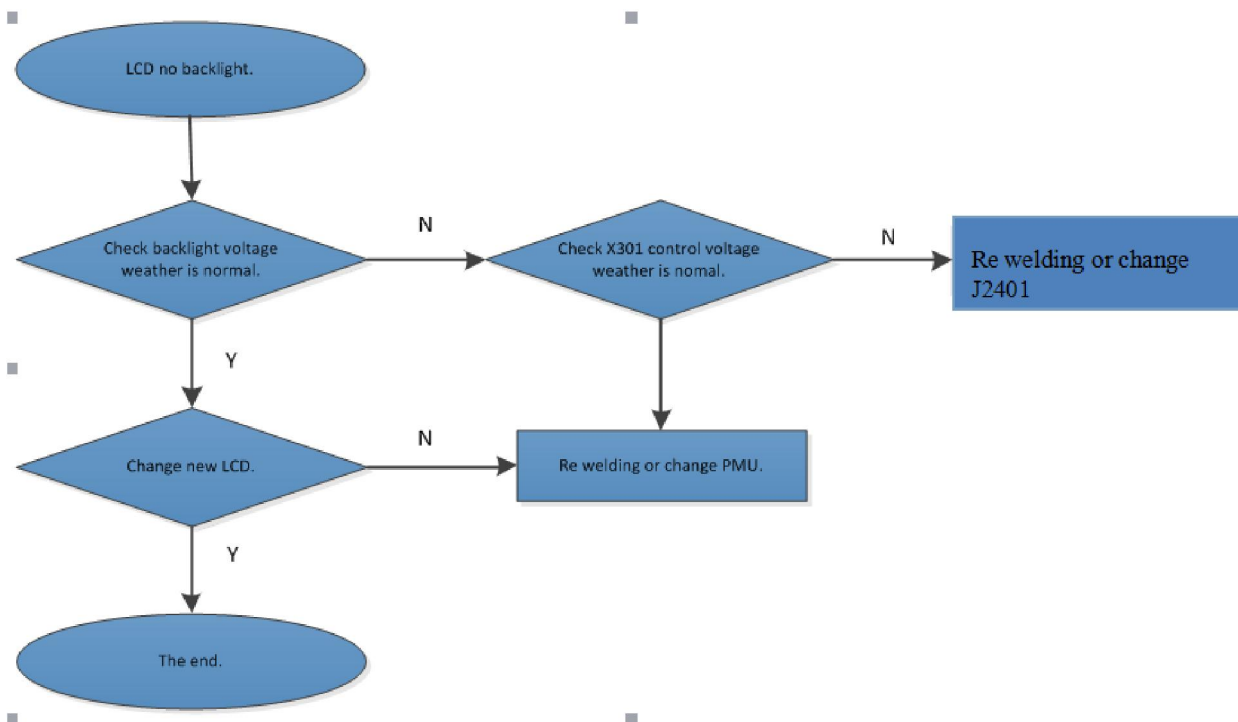
LCD blank screen and backlight broken:

Repair process: boot and test U2401 output VLED_P whether it is high, the signal for LCD backlighting lamp anode, LCD normal lit VLED_P is high, and check LCD welding is good; inspection VBOOST whether there is voltage (VBAT on this signal is received), Check if LCD_BL_EN is high.

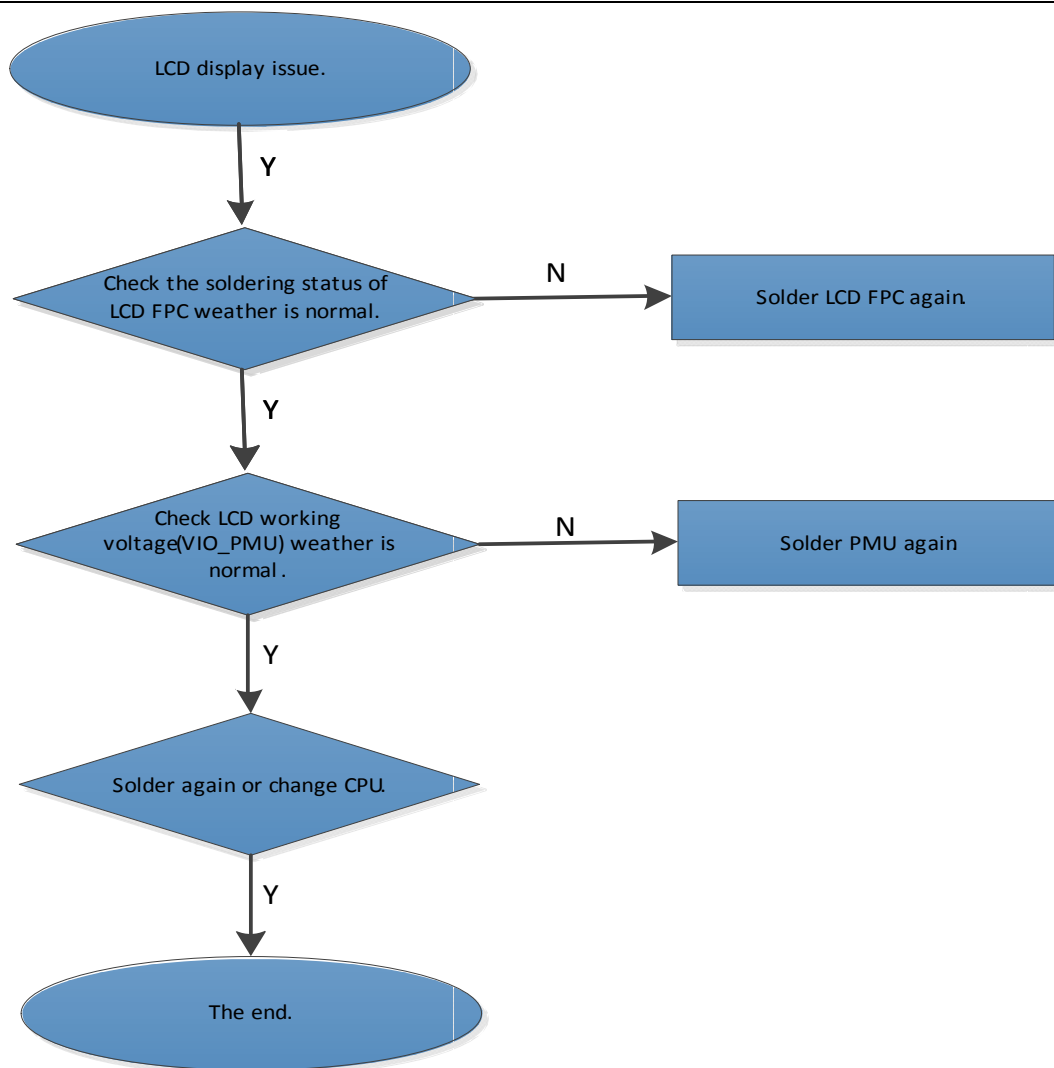
➤ Circuit Analysis:

The display of ZC600KL used 4.5 inch LCD. Its data interface include MIP Iport and signal controlled lines, The digital port of LCD is driven by BBIC MSM8937, and its power supplied by VIO 2.8V of PMU IC, and its driven circuit of backlight is supplied by the four comm onanode and backlight cycles of J1705.

LCD backlight issue repair flow:

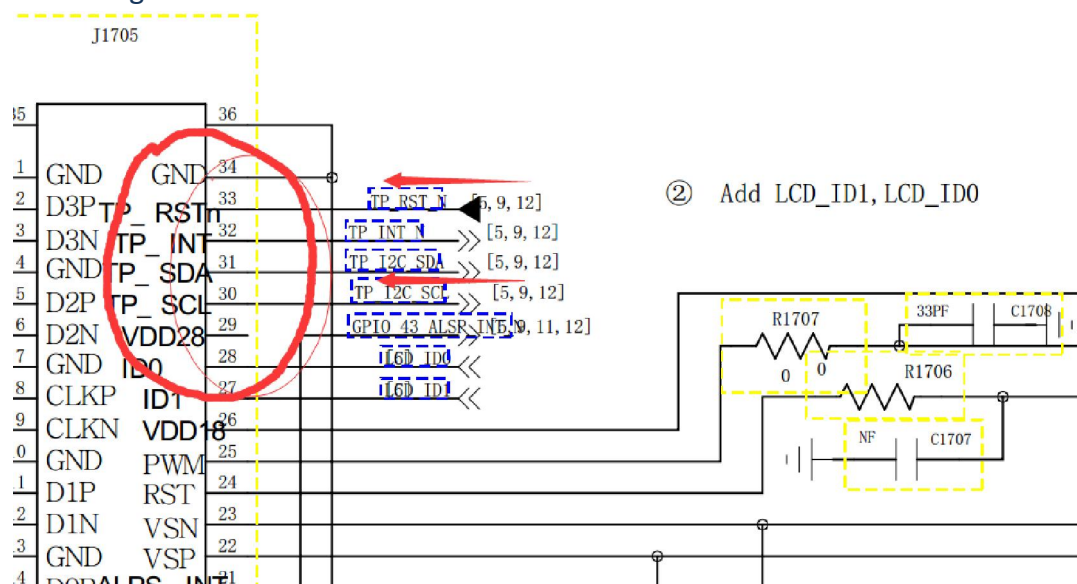


LCD display issue repair flow :



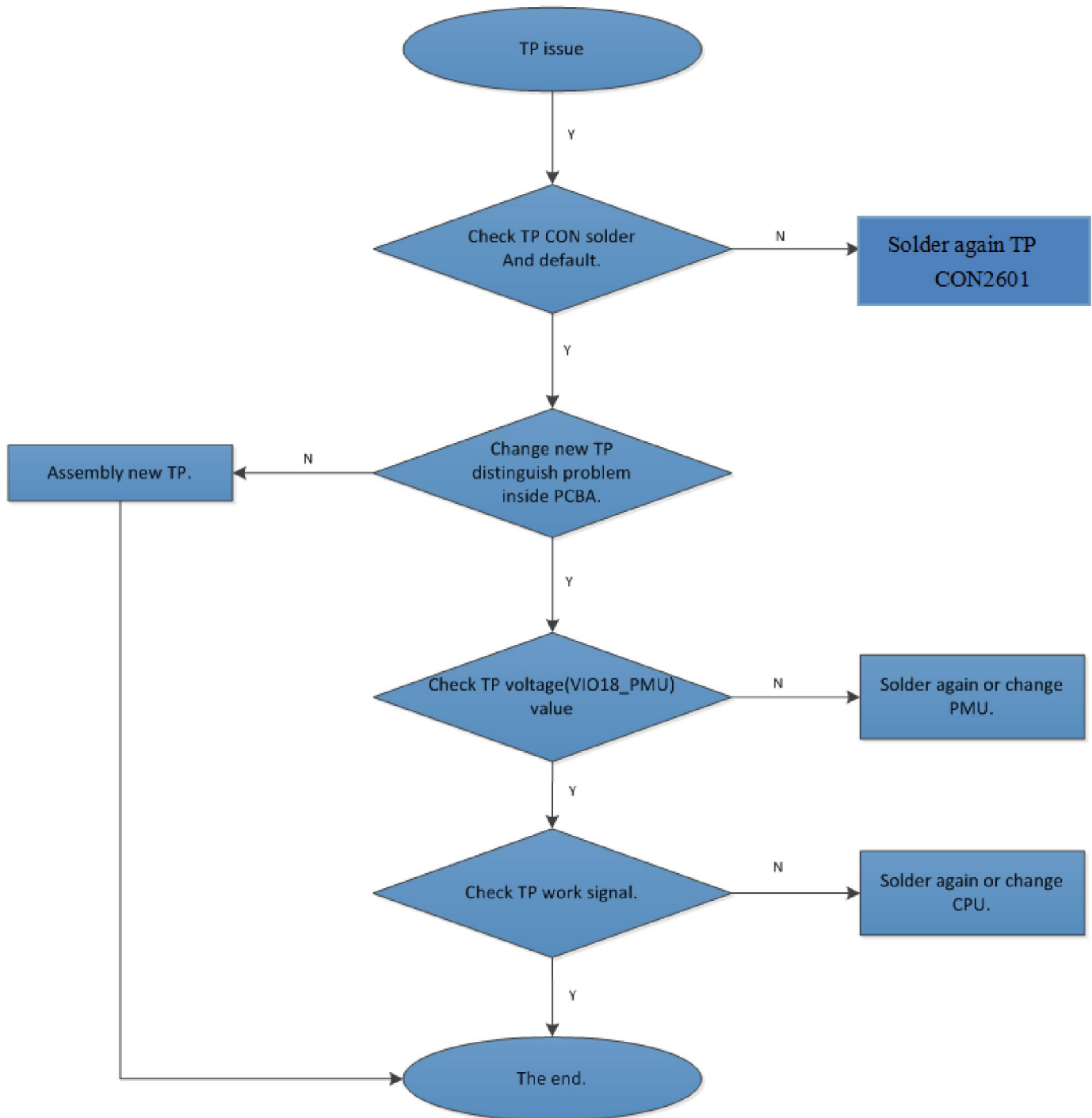
3: Touch panel

Block Diagram



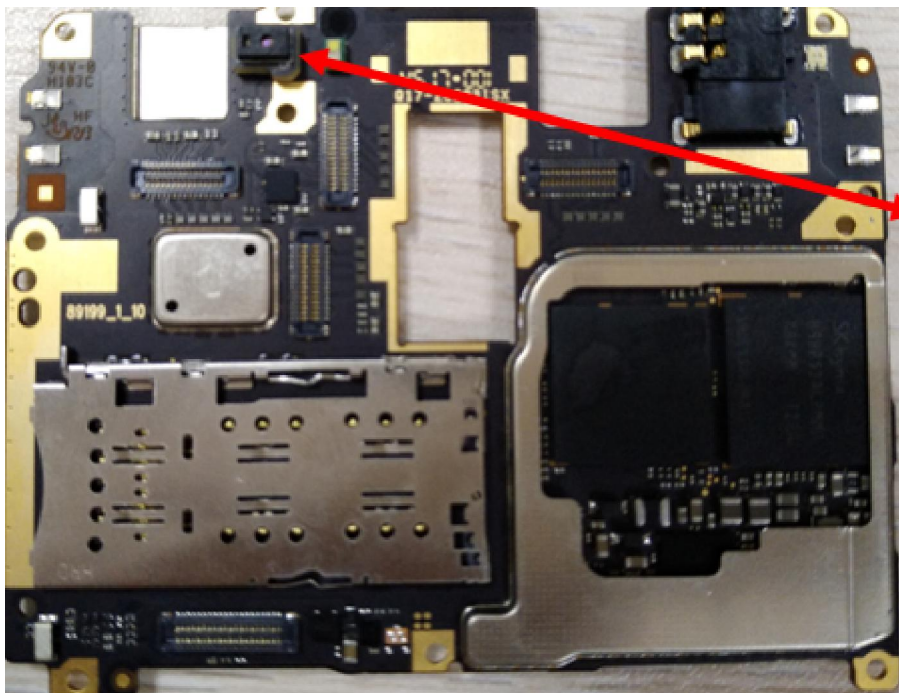
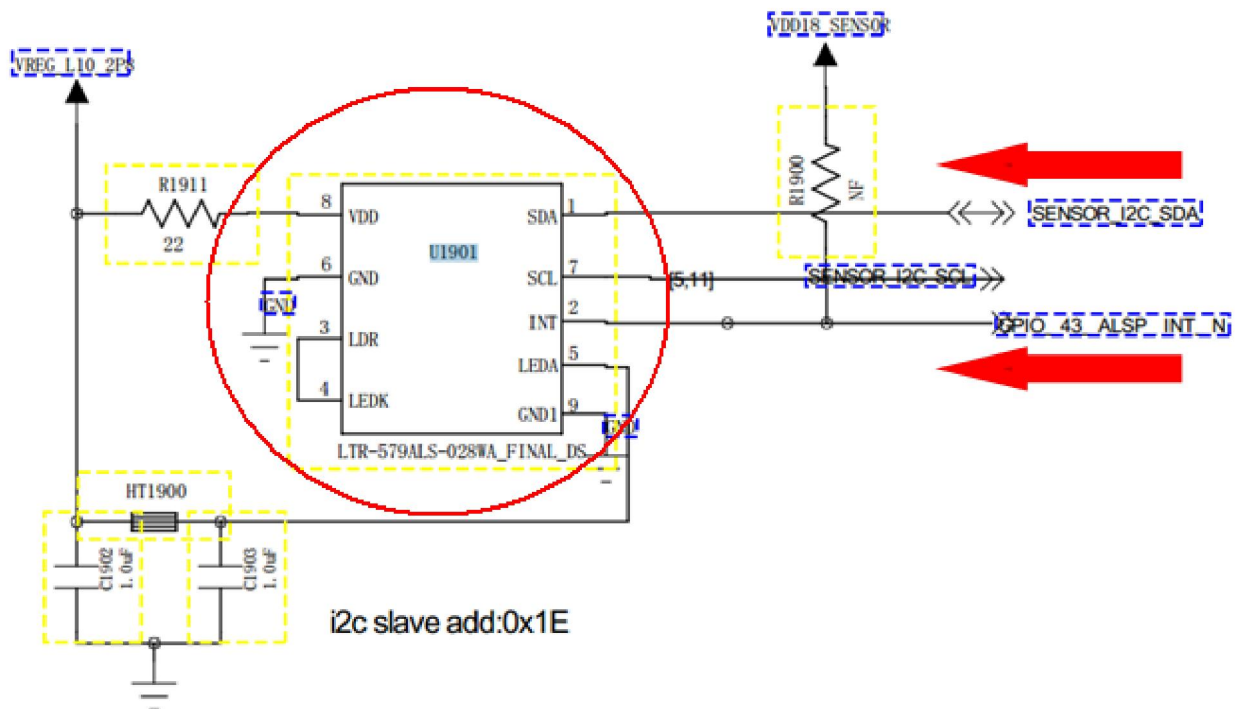
J1705
BTB-Connector
TP

4.TP Issue repair flow:



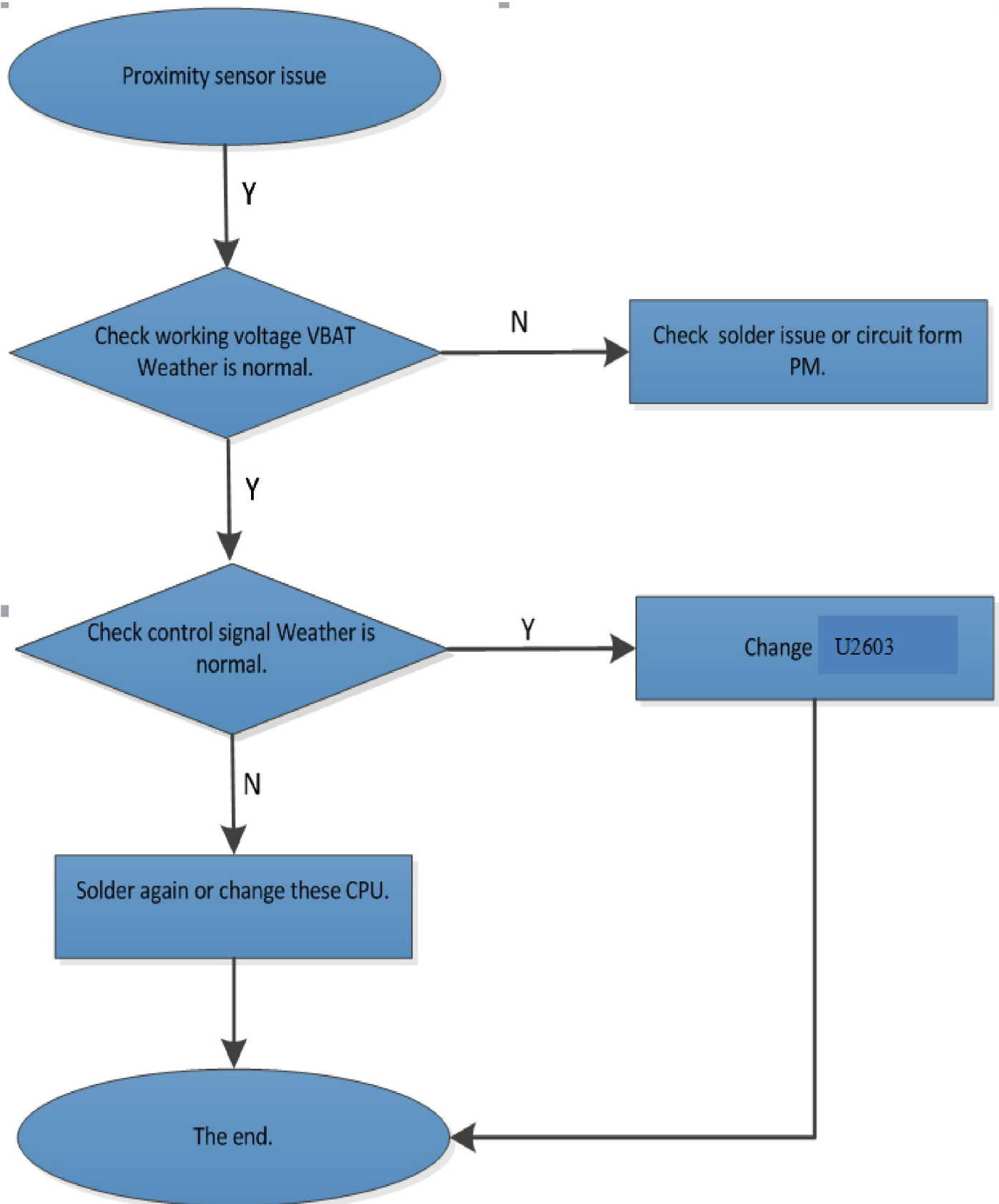
5: Proximity Sensor

Block Diagram



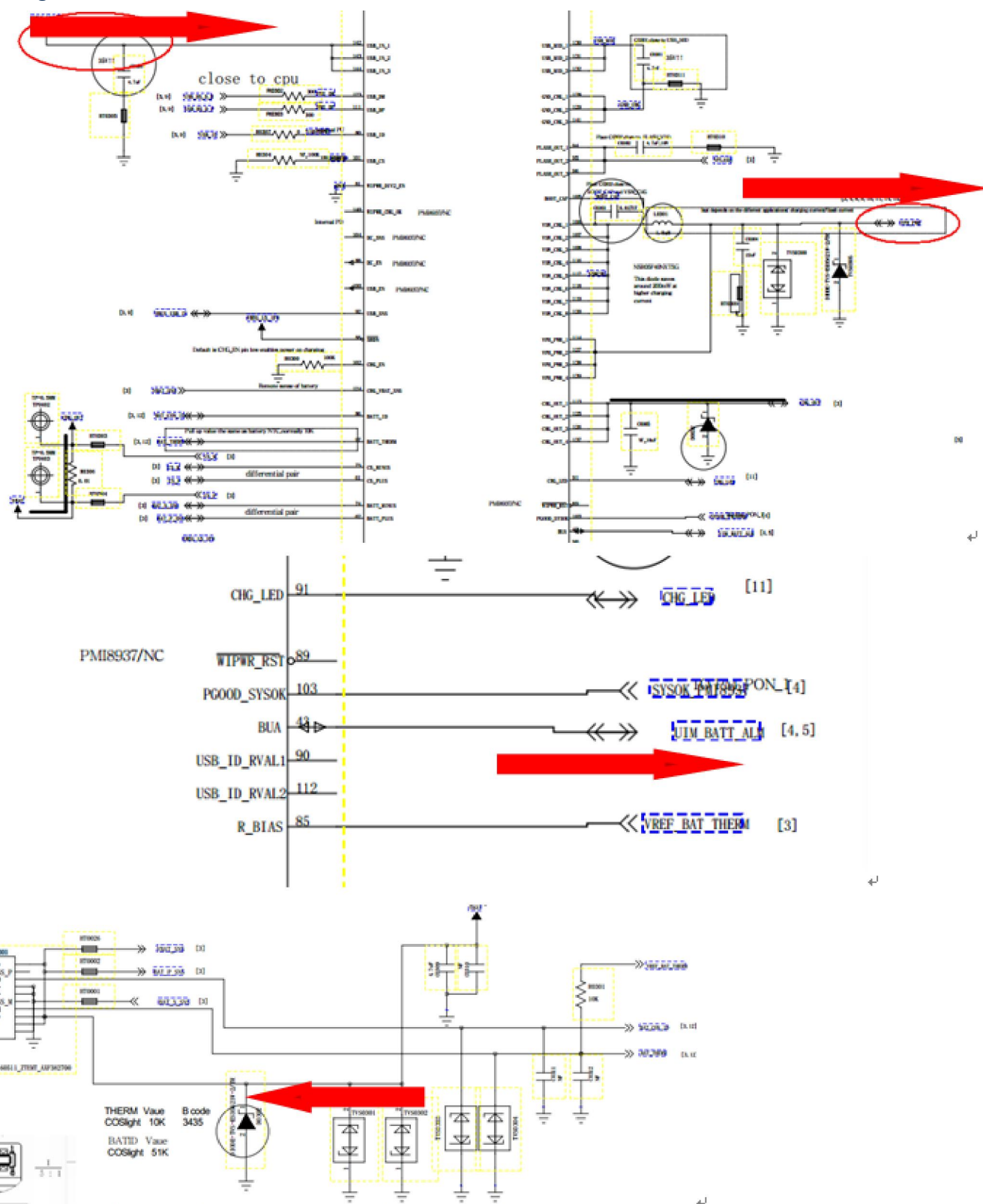
U1901+
Light-sensitive+

Proximity issue repair flow:



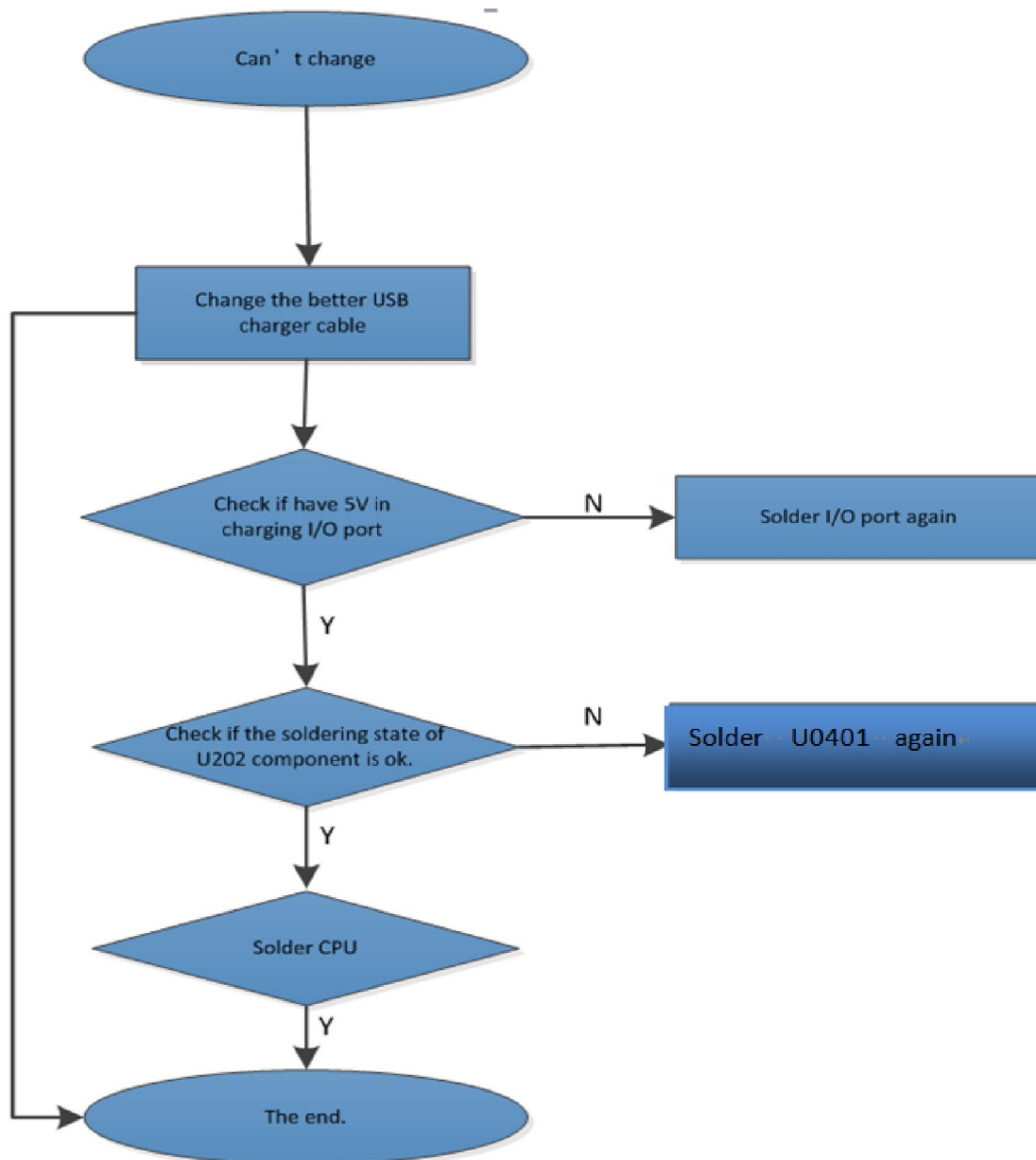
6: Charging

Block Diagram





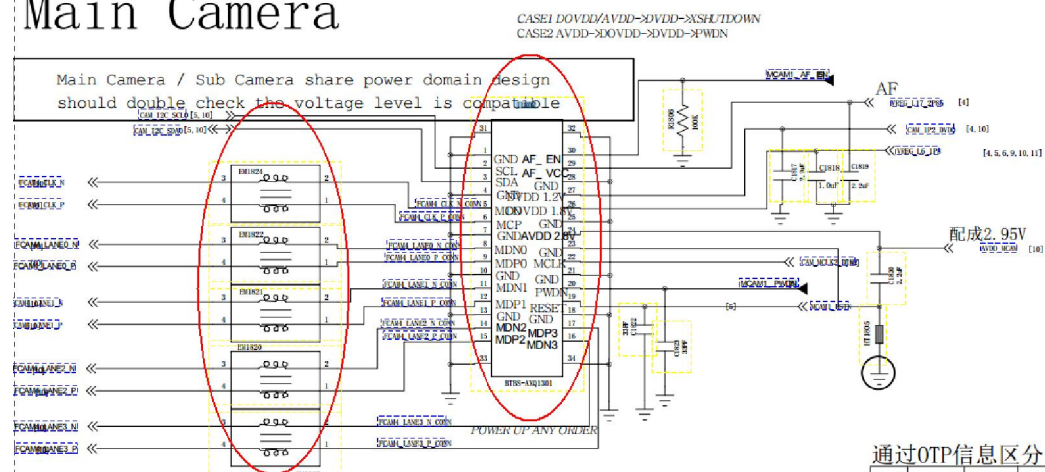
Can't charge repair flow:



7: Camera

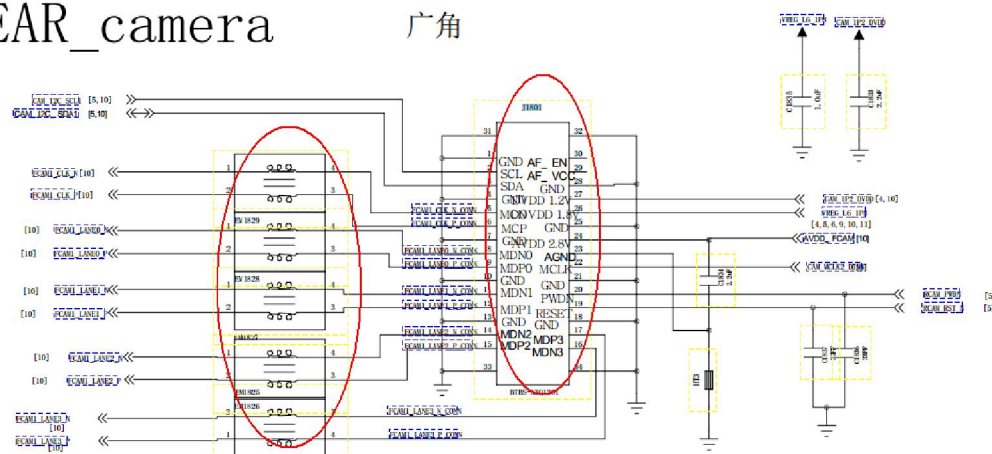
Block Diagram

Main Camera



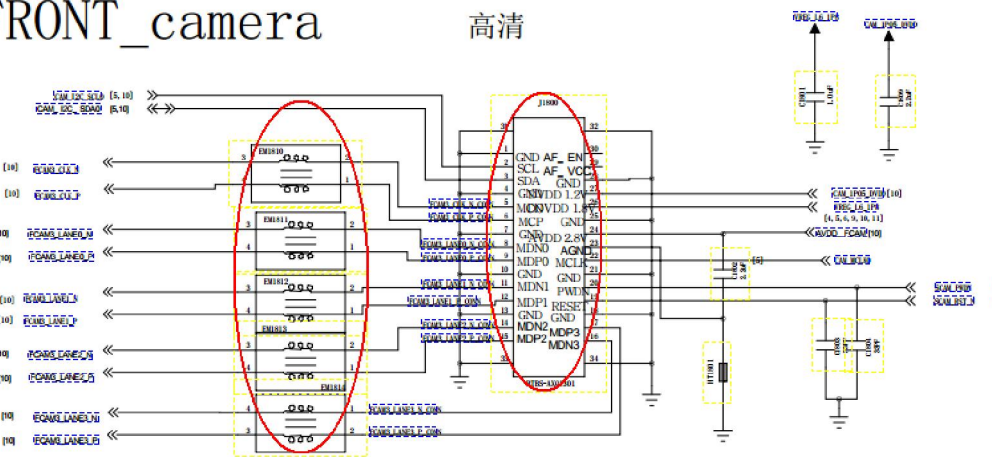
REAR_camera

广角

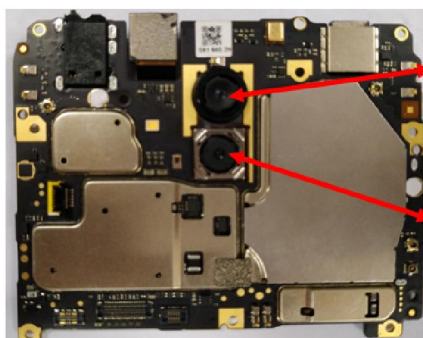
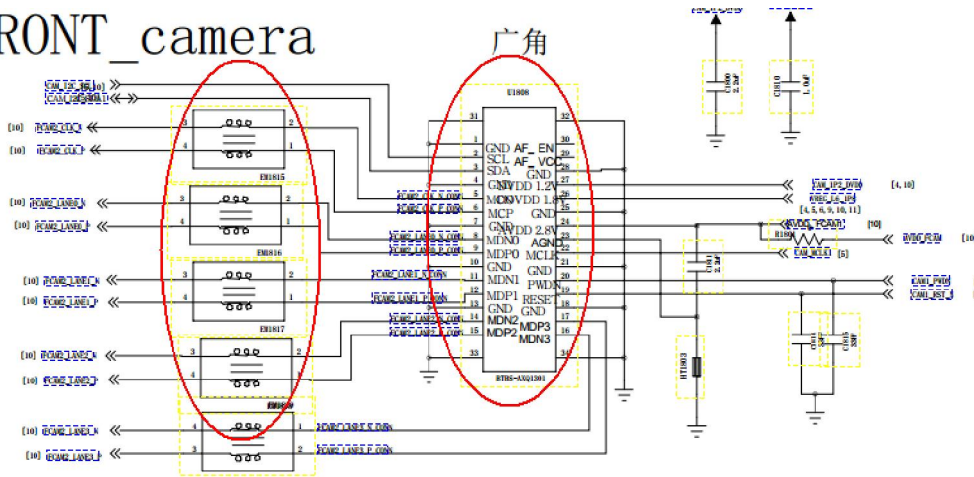


FRONT_camera

高清

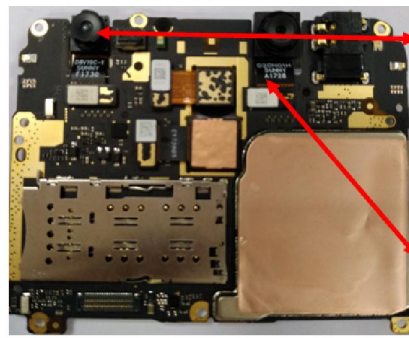


FRONT_camera



J1801 After the wide-angle camera

J1802 The main camera



J1808 front camera

J1800 Wide-angle front camera

Failure analysis is as follows:

After the replacement camera, a camera to eliminate ontology failure problem; Check the connector J1401/J1402 welding problem again.

So camera function could not be opened:

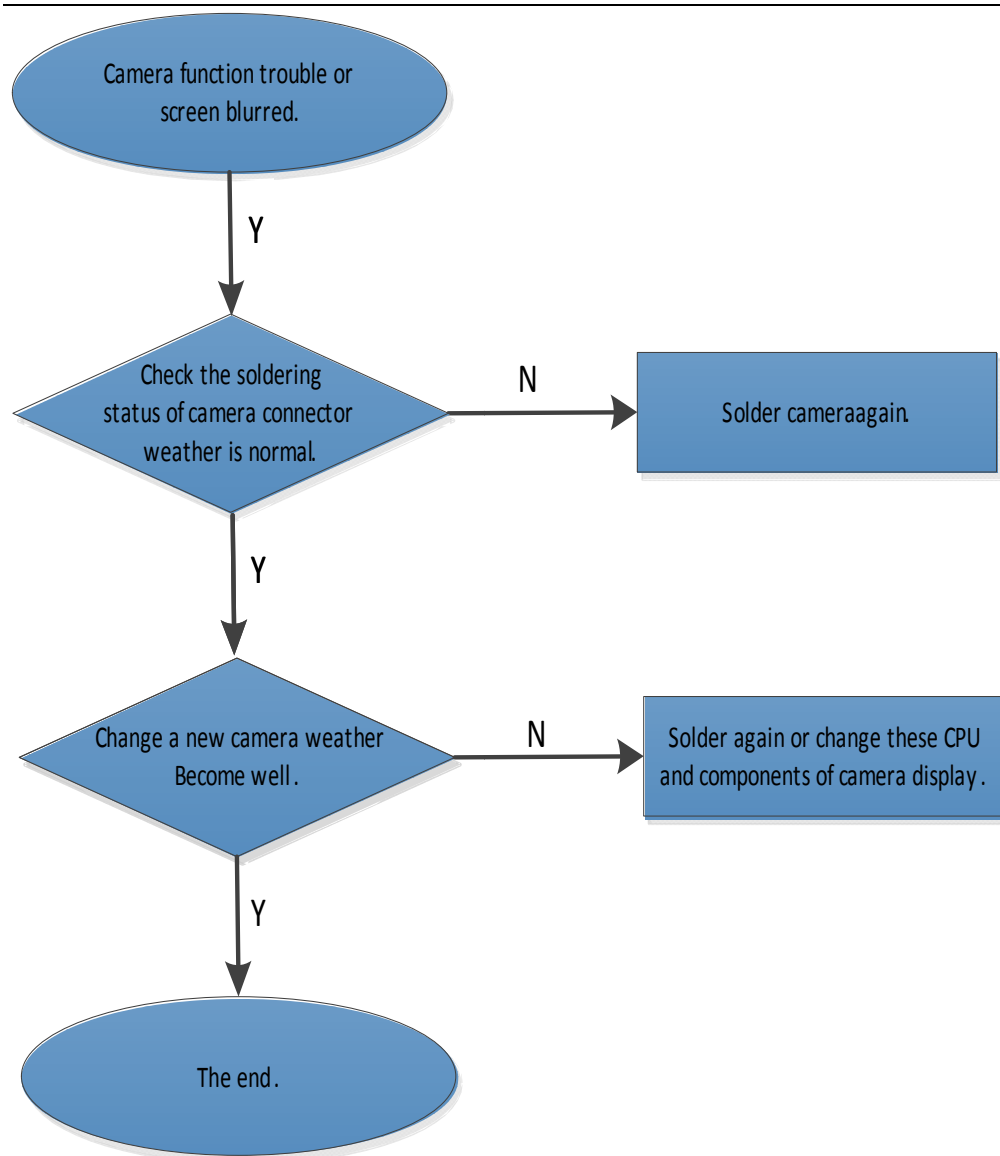
1. Check the power supply voltage 'VDDCAMMOT', 'VDDCAMCORE', 'VDDCAMIO', is normal, such as abnormal change - PMU - 2.
2. Review the control signal of cameras' CMMCLK ', 'REAR_CAM_ID0', 'CCIR_RET', voltage and the circuit is normal.

All the examination without exception are to add welding/change the CPU.

So camera image display abnormal:

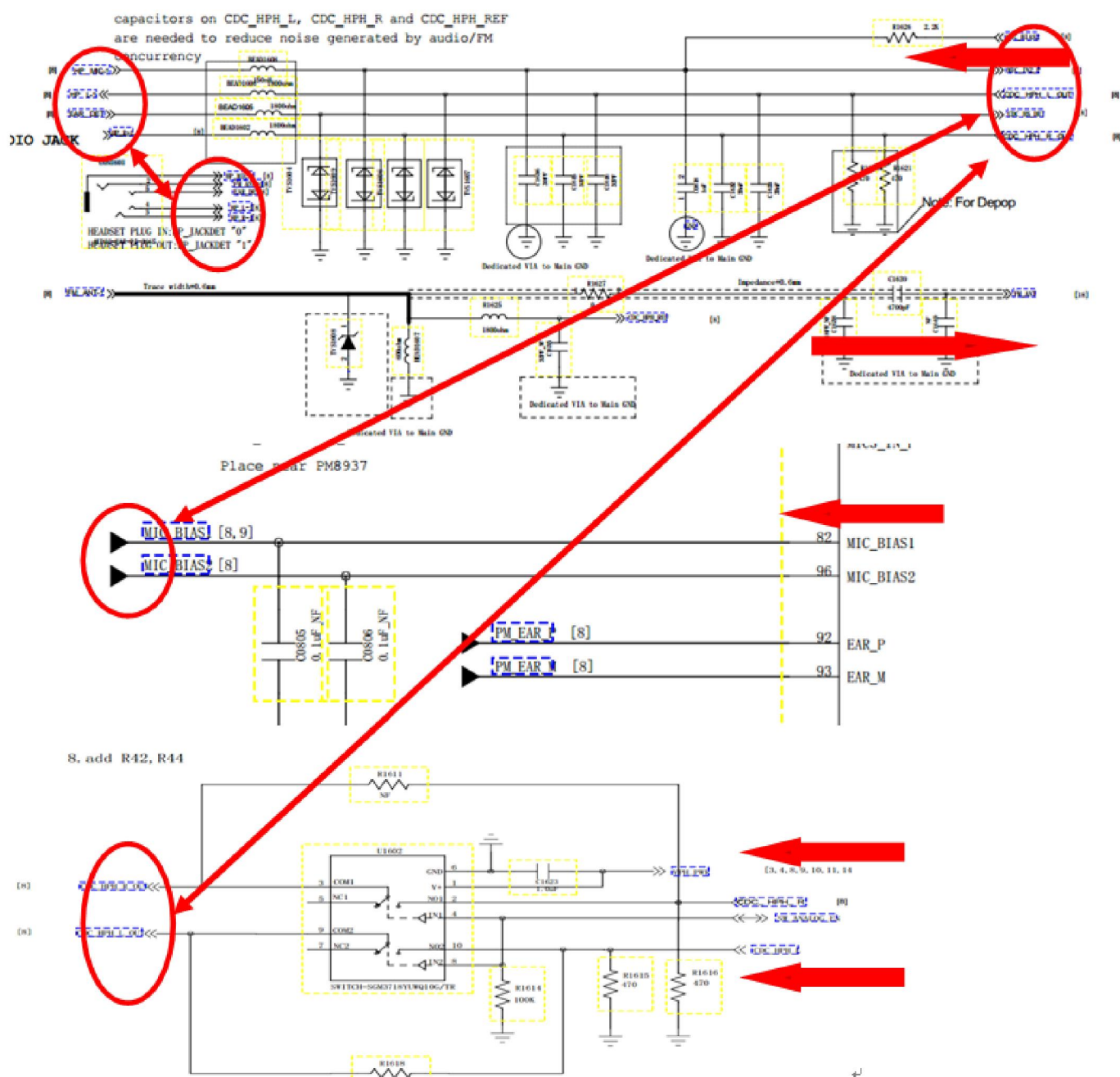
Check camera and transmission of data between the CPU signaling pathways conduction, whether there are abnormal check pathway on the EMI0810-0825 welding device and the performance is normal, if there is no exception to welding the CPU.

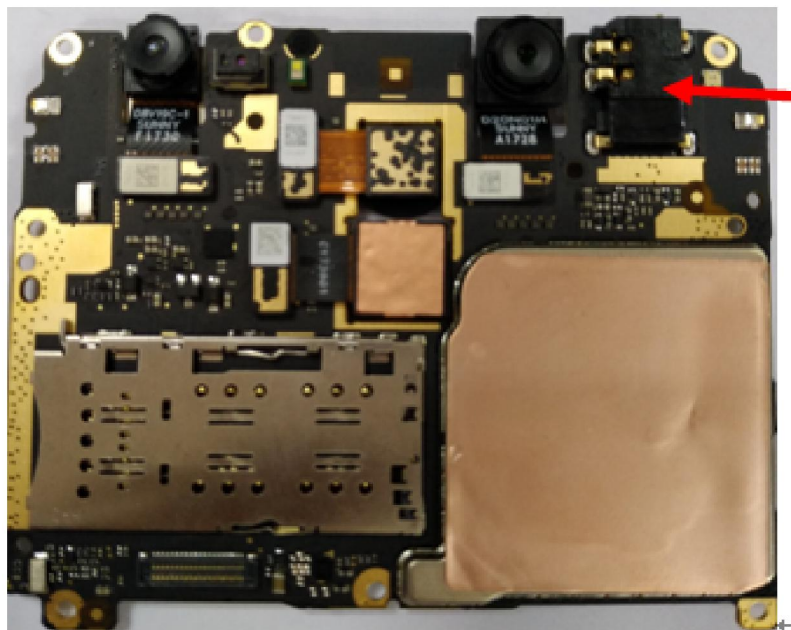
Camera issue repair flow:



8: Voice issue

Block Diagram





• **CON1601**
Headphones-sit

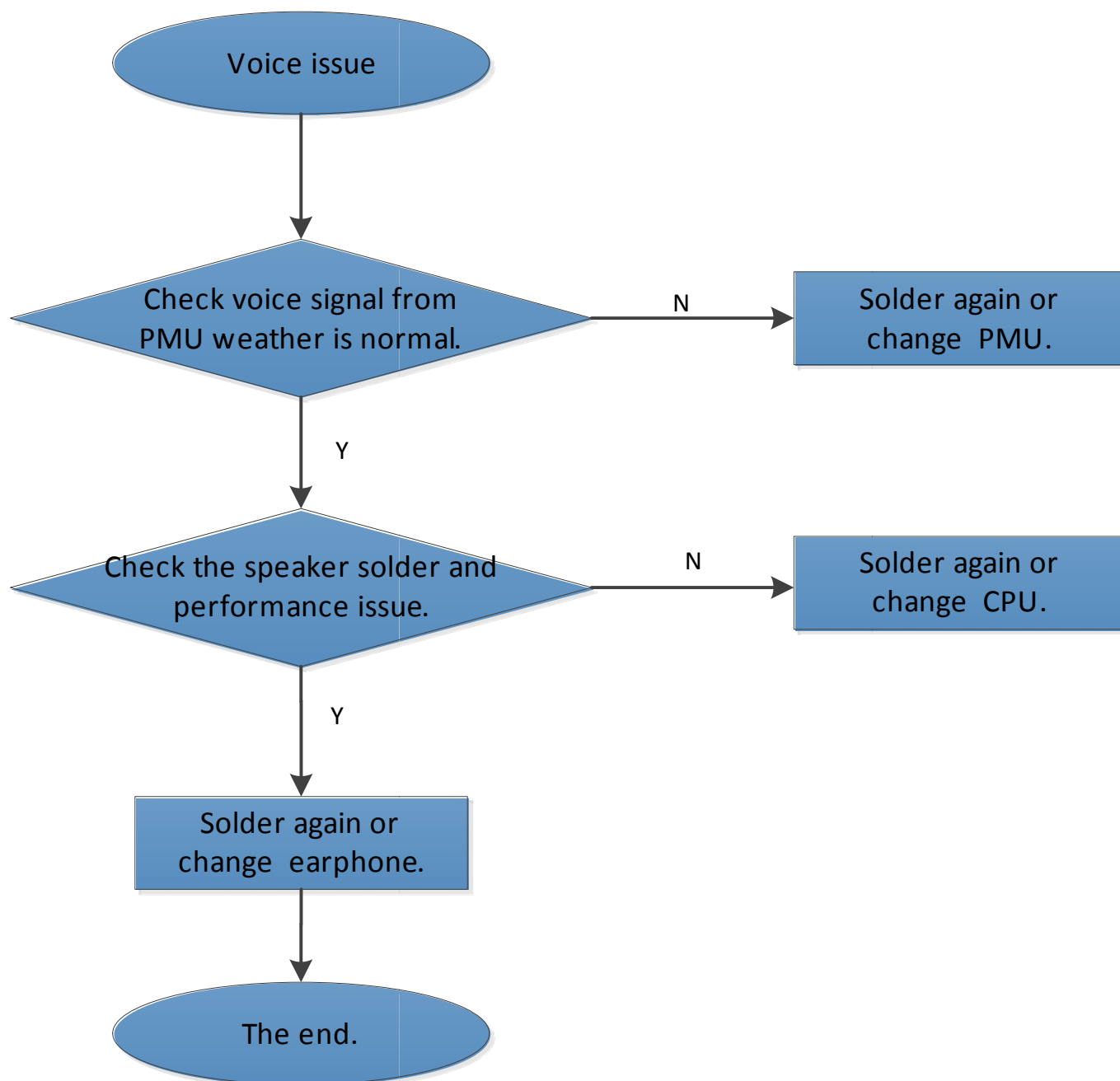
General analysis process is as follows:

1. The measurement of the receiver are the resistance between the two pins for 28 Ω , such as abnormal test again after replacing a receiver, to rule out the failure of the phone itself.

2. Test the receiver input signals' EARP ', 'EARN' voltage is normal, such as abnormal first check welding line on the device and the performance is normal, especially electrostatic protection tube D3103 (can disconnect first to determine whether the signal attenuation).

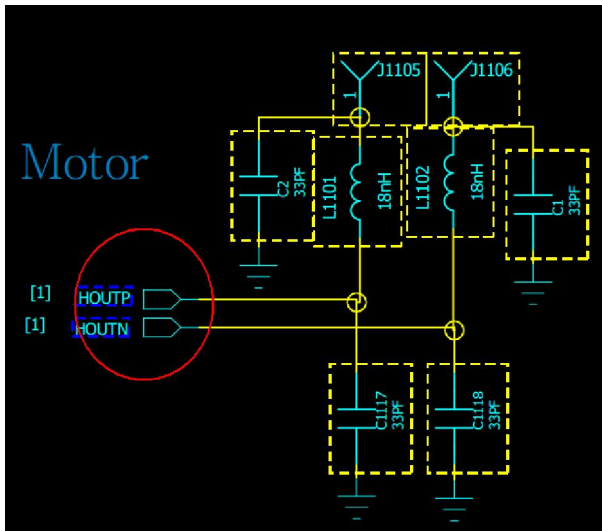
If the above are no obvious problems, check the CPU or to replace it.

Voice issue repair flow:



9: Vibrator

Block Diagram

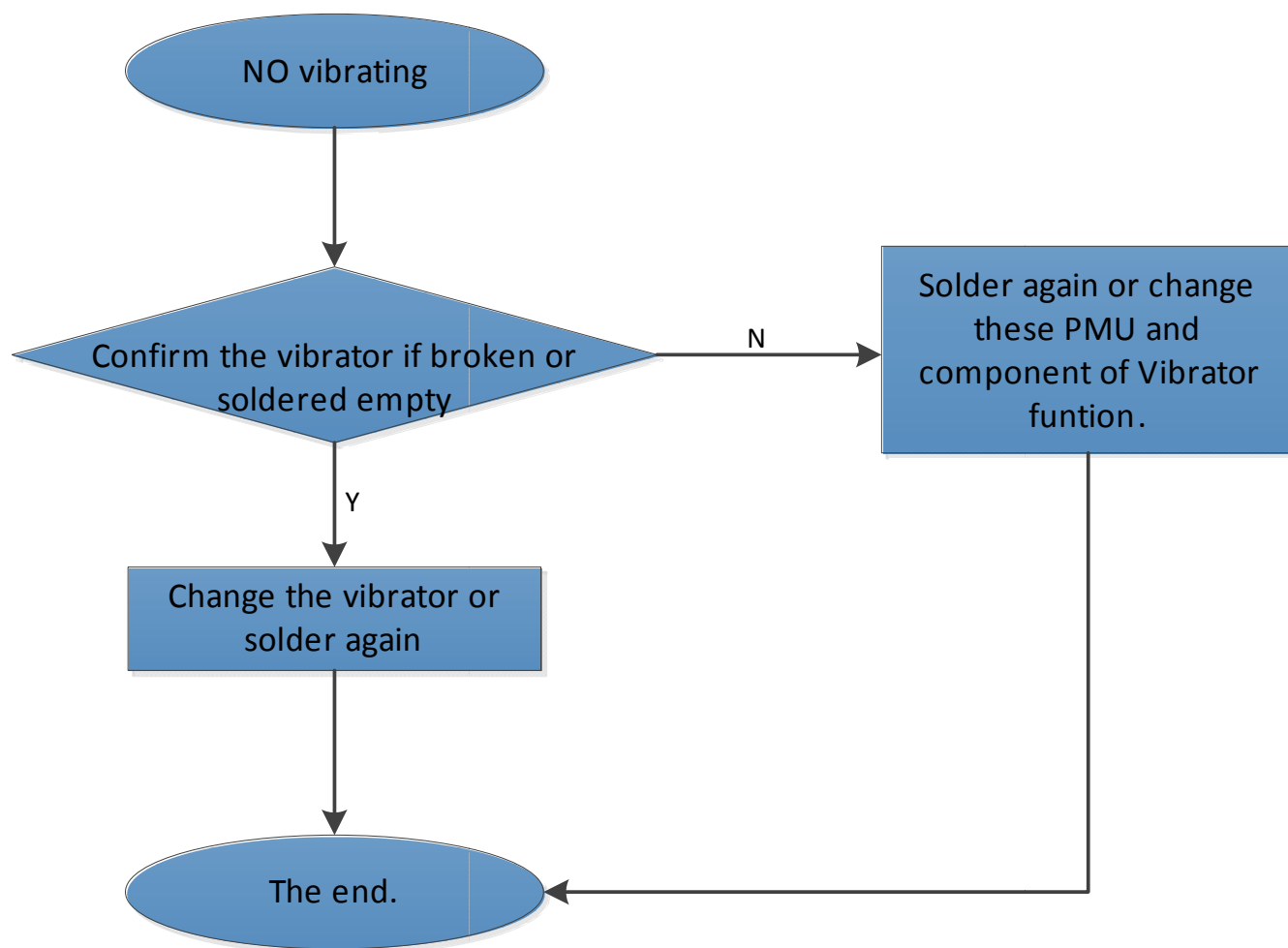


motor

General analysis process is as follows:

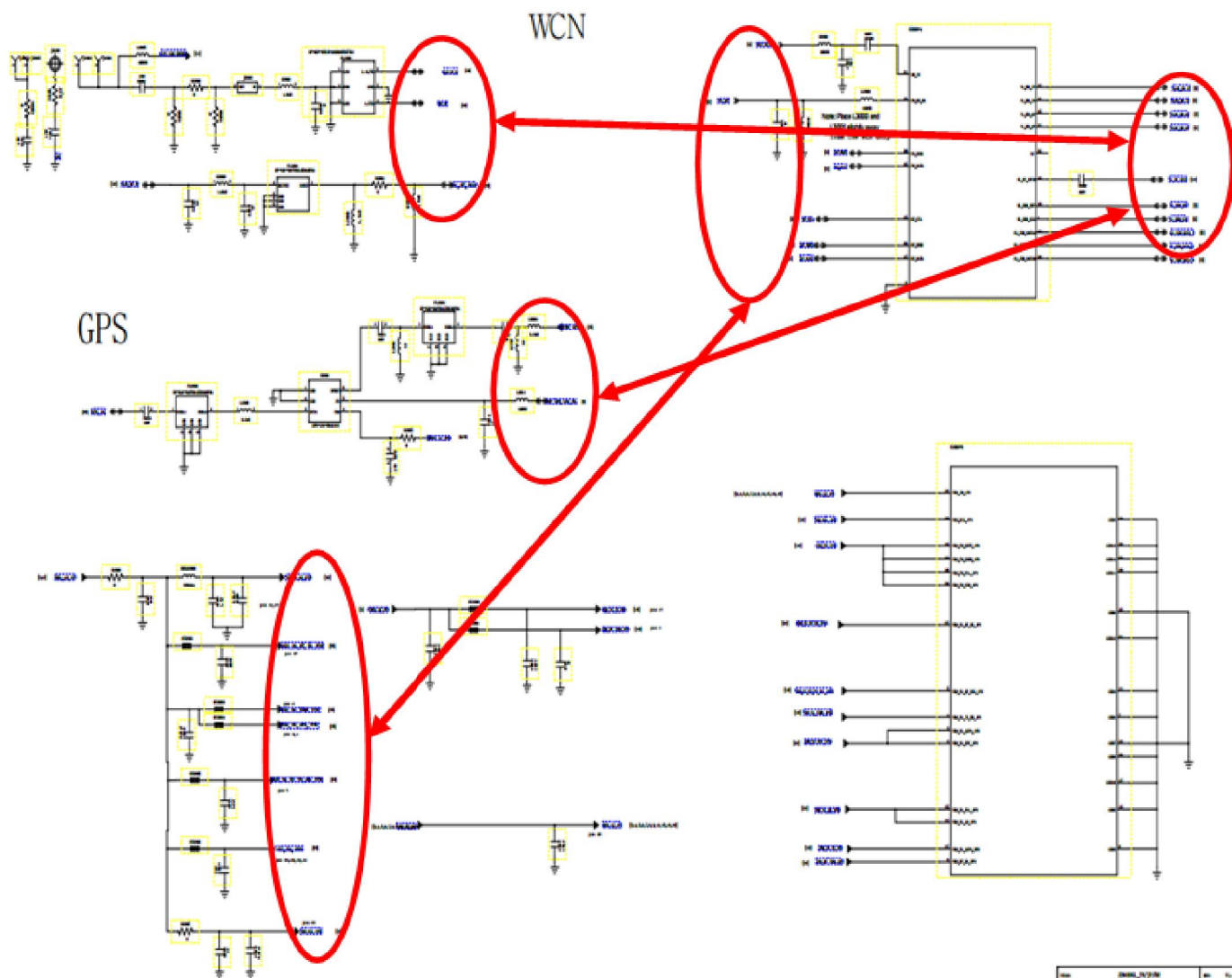
1. Check if motor is negative resistance between 33 Ω , such as abnormal need to be replaced; Check the motor of the welding is reliable.
 2. Check whether the motor working voltage 'VIBR_PMU' normal loaded into the small plate 'VIB_CTRL' this point, the motor whether the cathode and the mainboard of ground conduction;
- If the above are no obvious problems, check - PMU - 2, or to replace it.

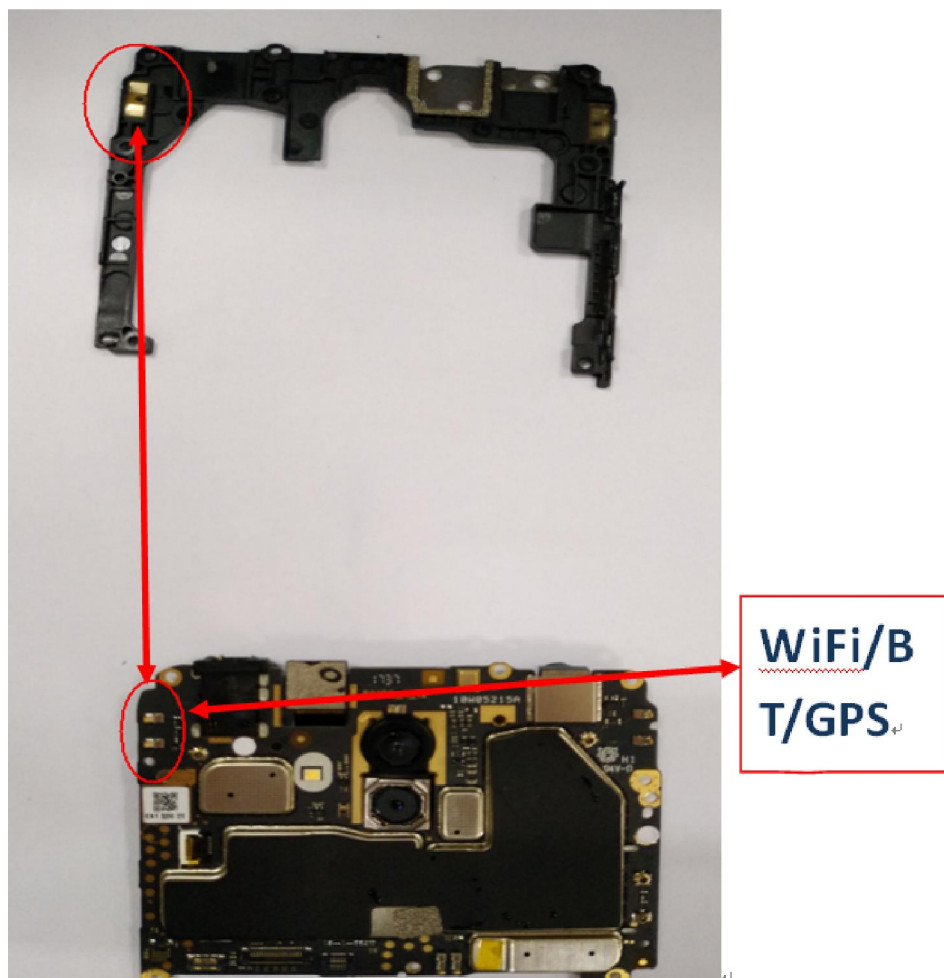
Vibrato issue repair flow:



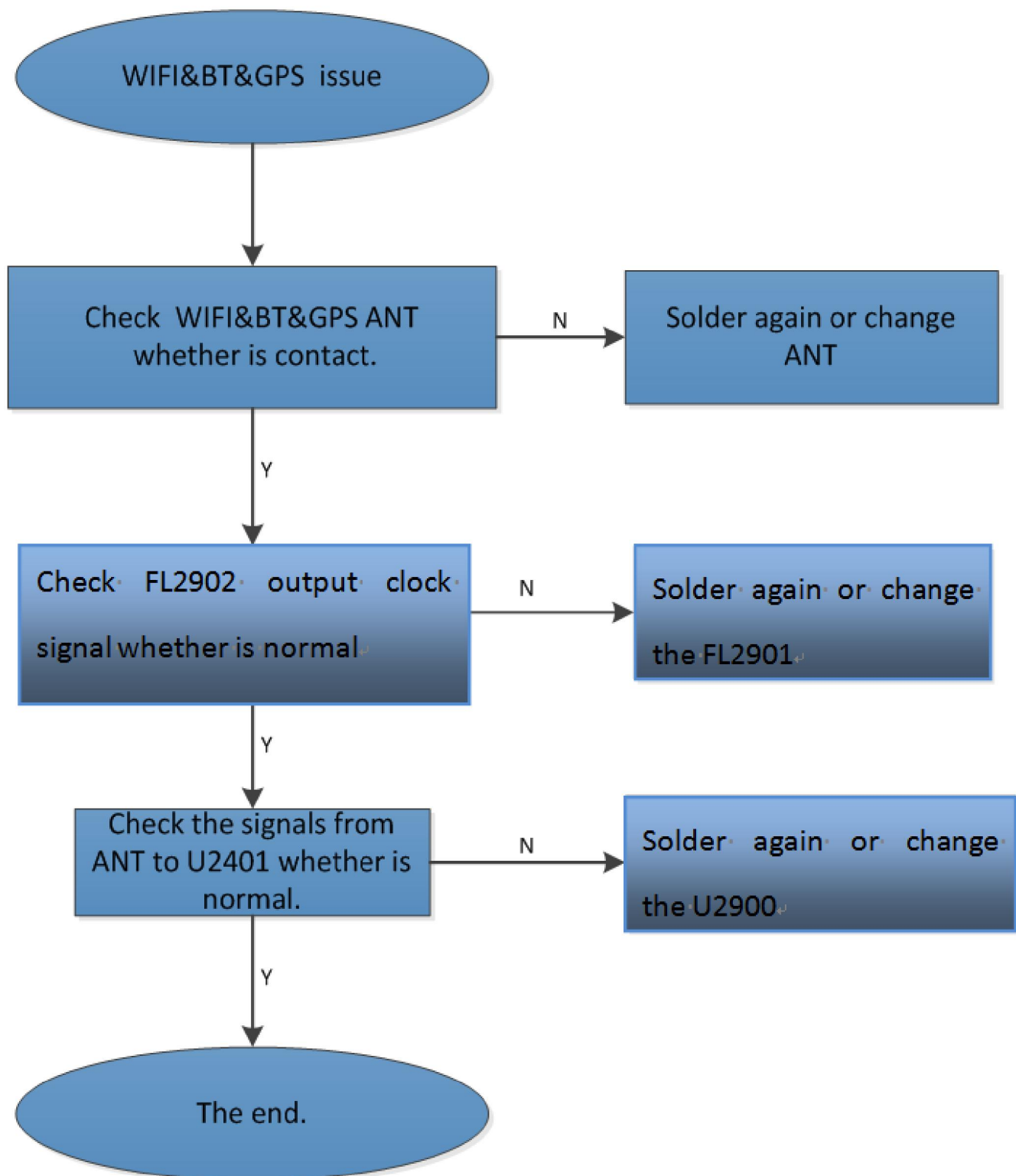
10: WiFi/BT/GPS

Block Diagram



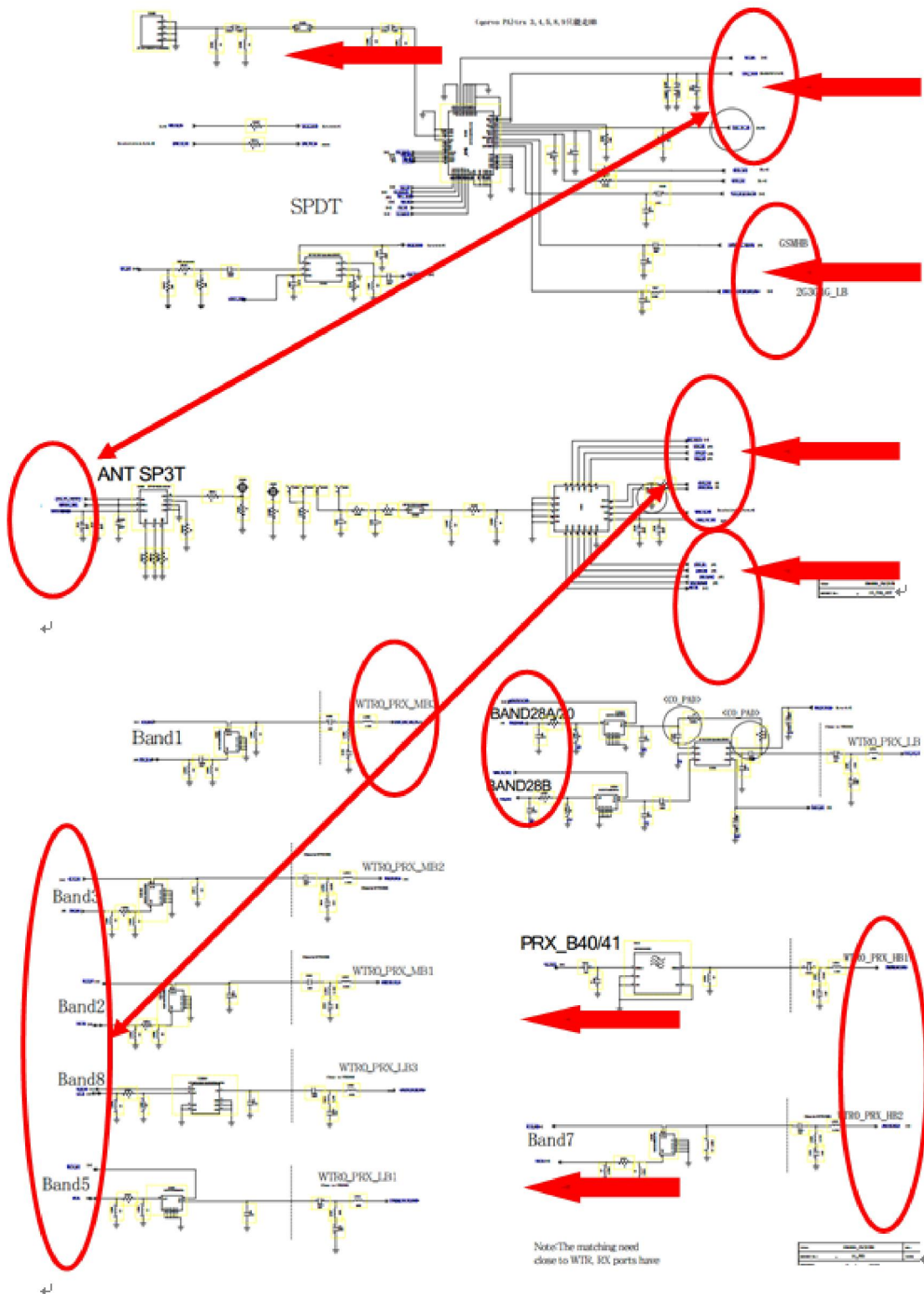


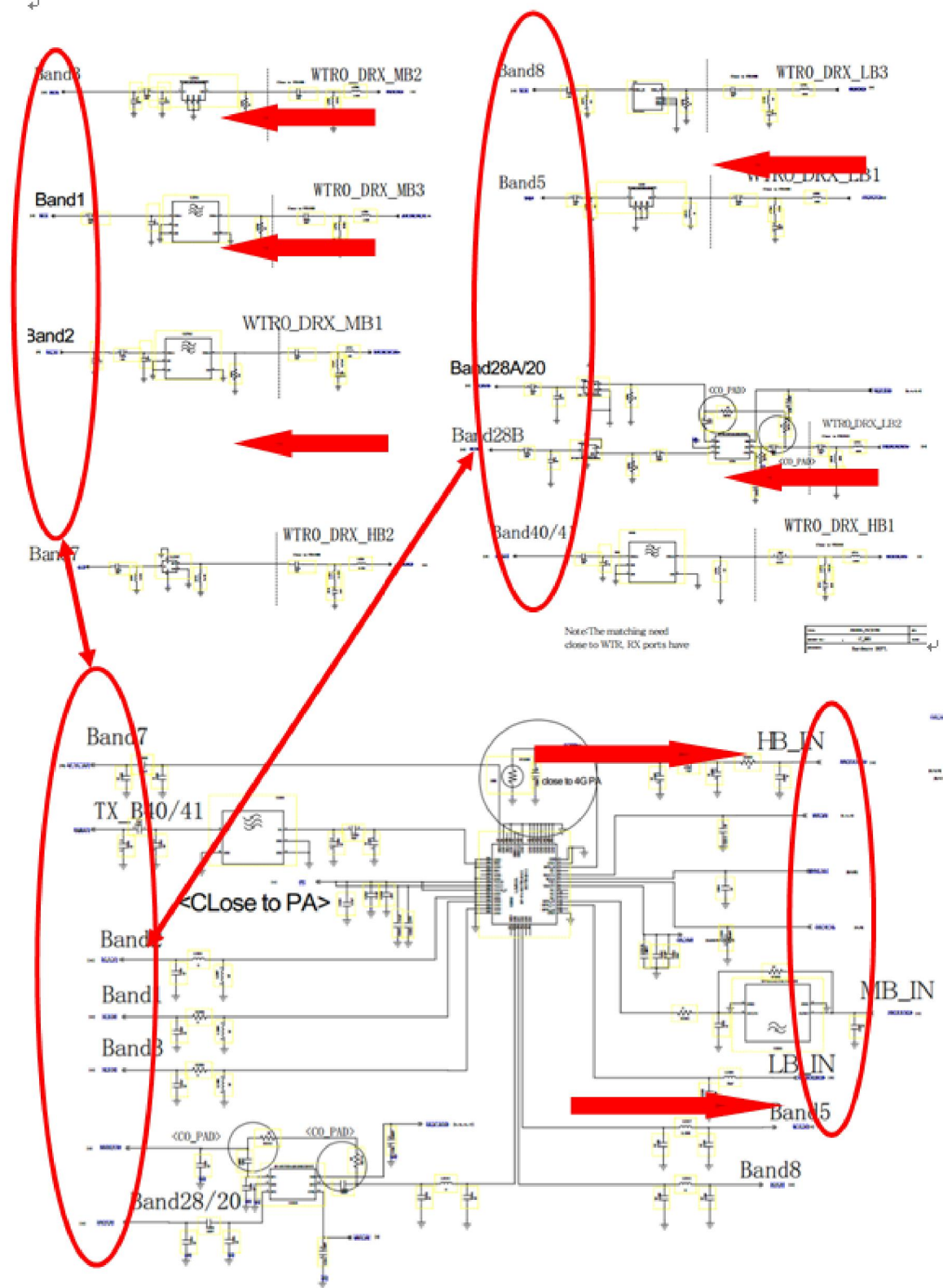
Wifi&BT&GPS Issue repair flow:



11: RF Network

Block Diagram



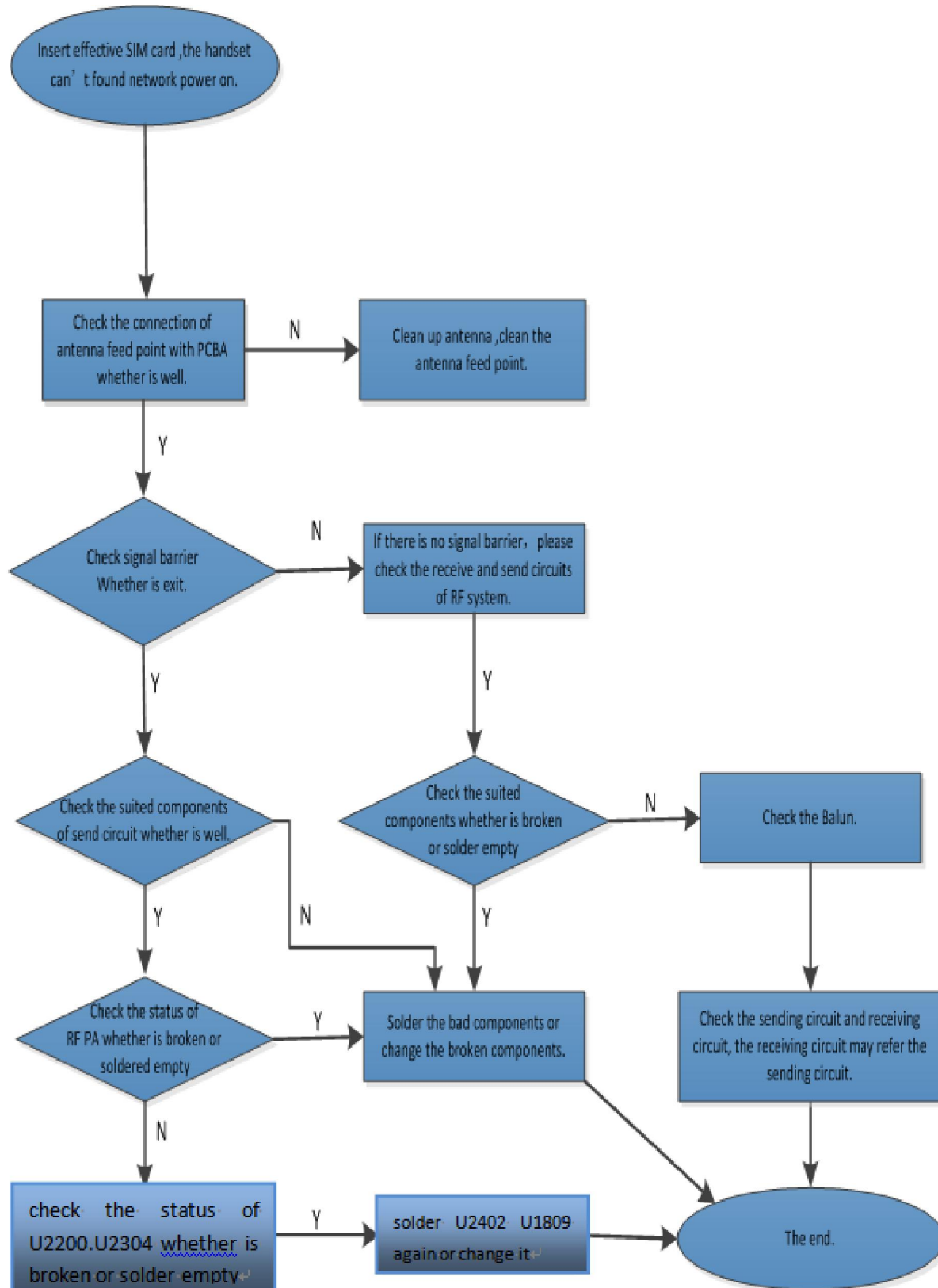




RF head
interface



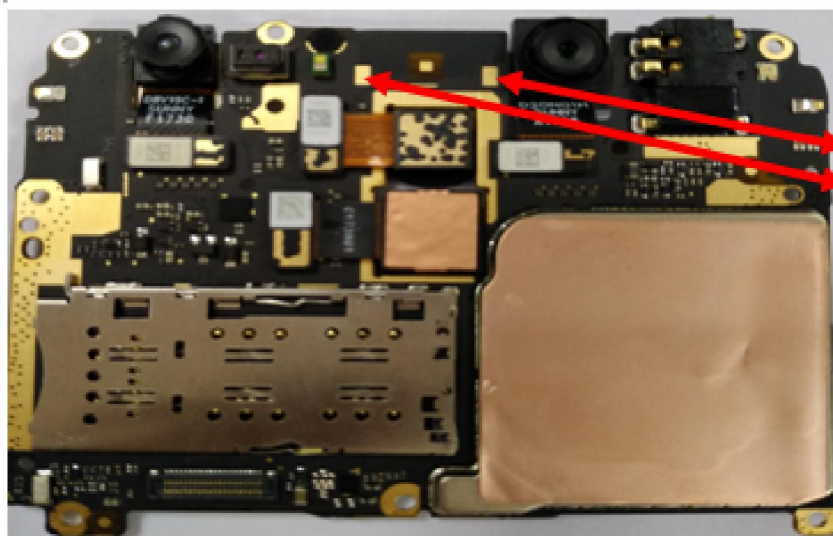
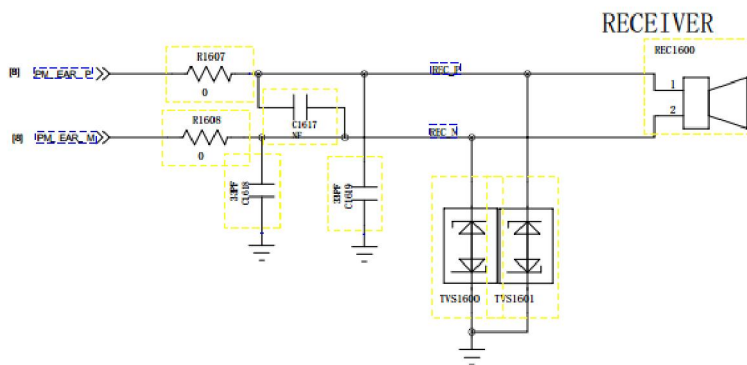
RF head
interface



RF Issue repair flow:

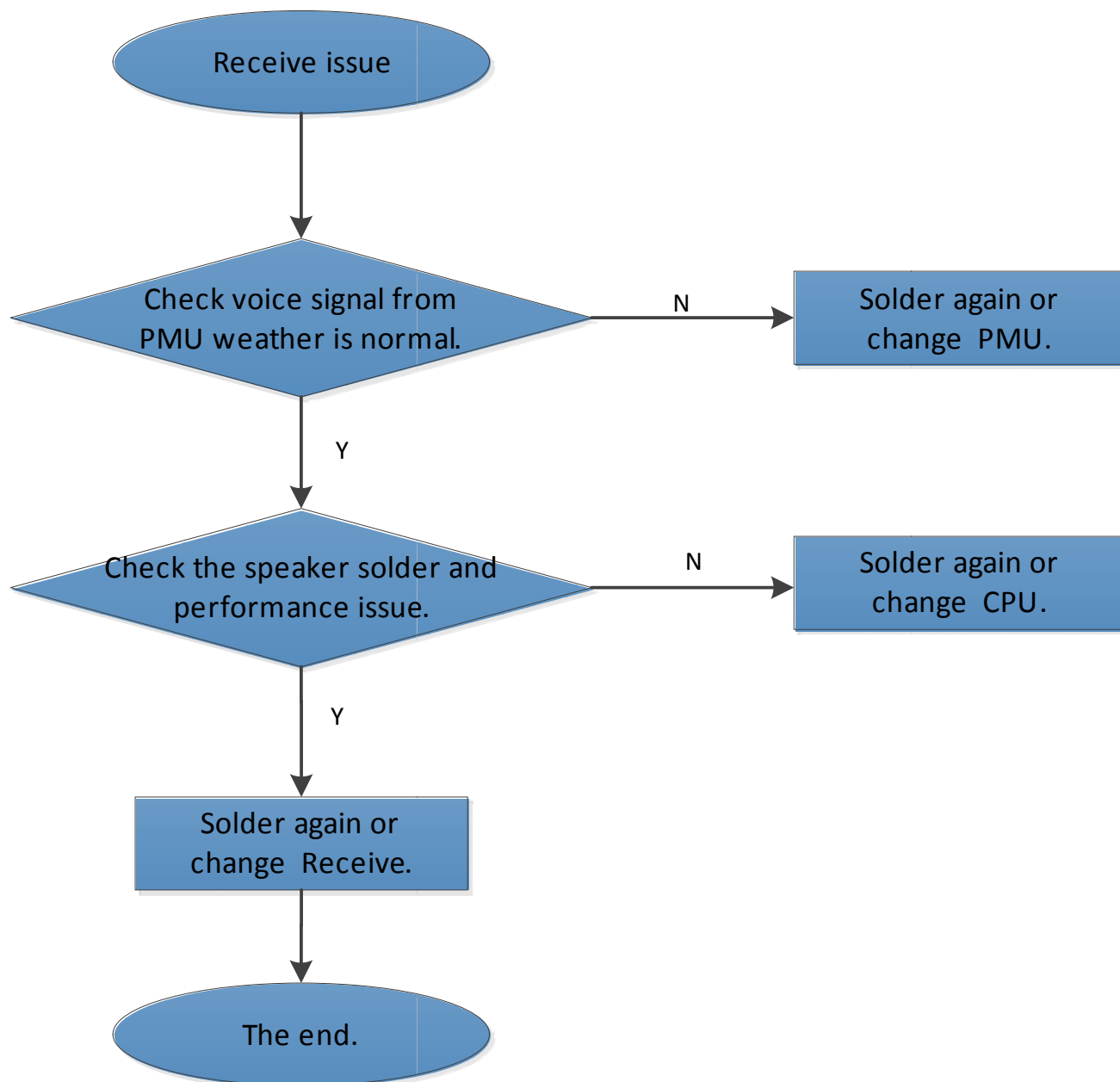
12: Receiver

Block Diagram



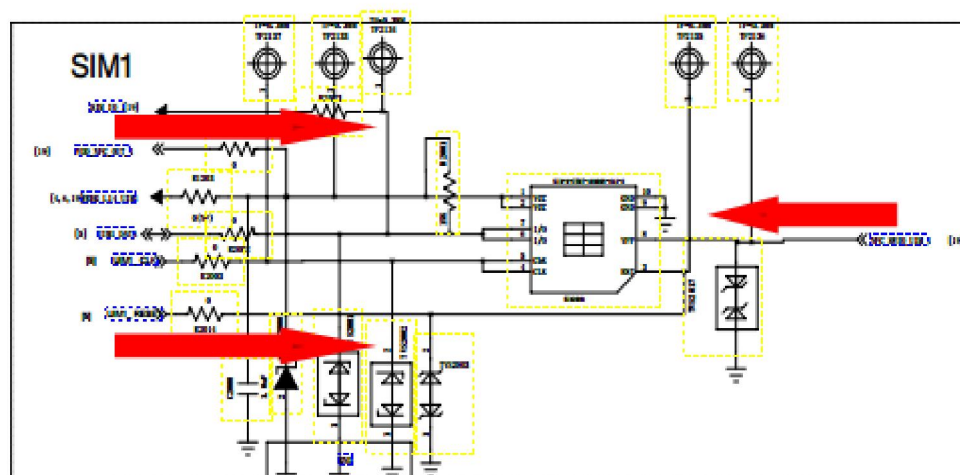
receiver

Receive Issue repair flow:

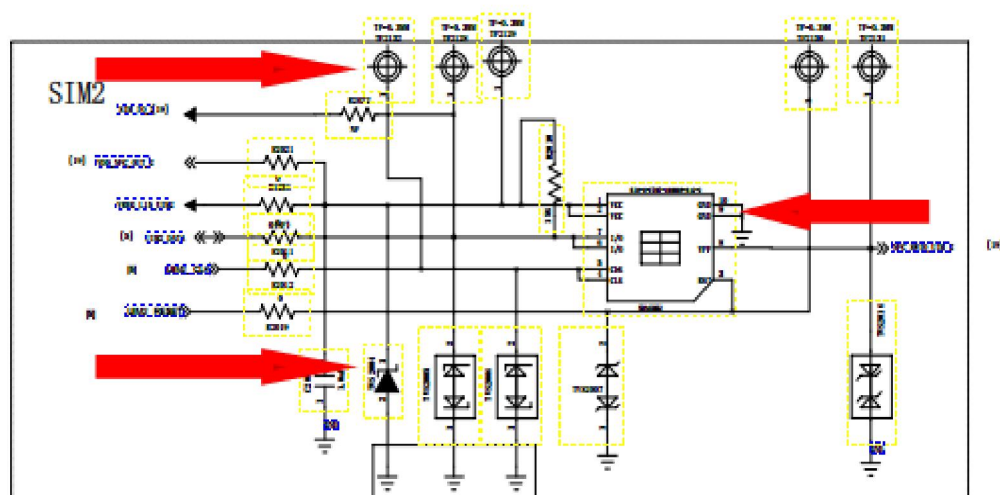


13: SIM card issue

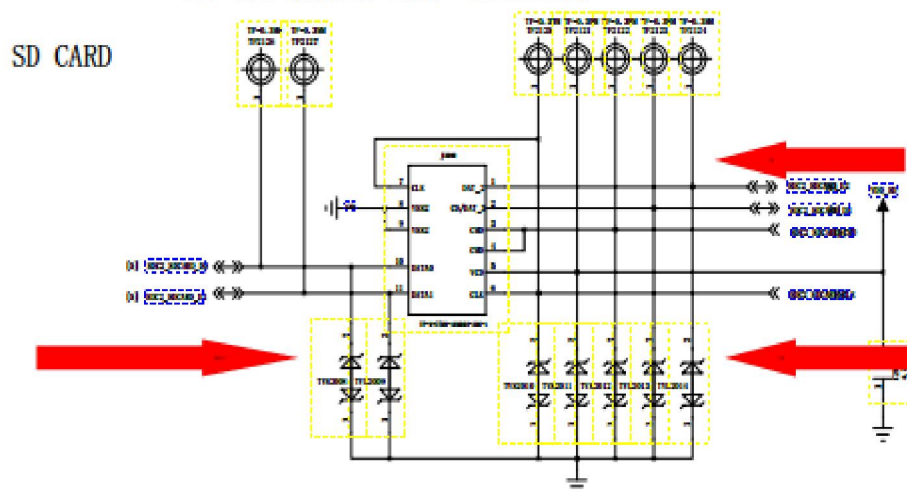
Block Diagram

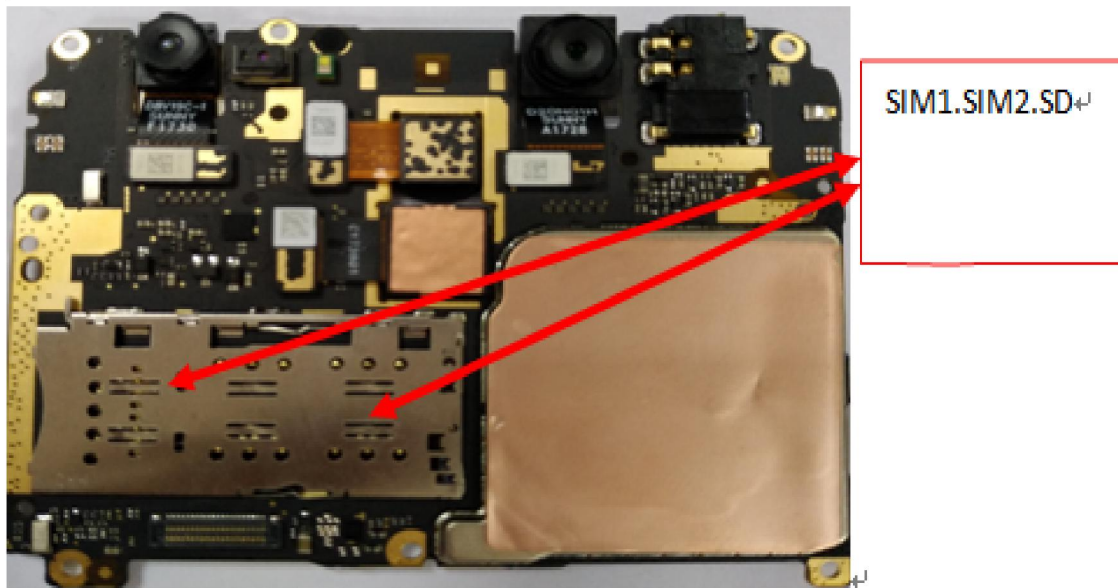


因为EOS测试, 请打孔下主地, 以有利于EOS最短路径泄放到地。



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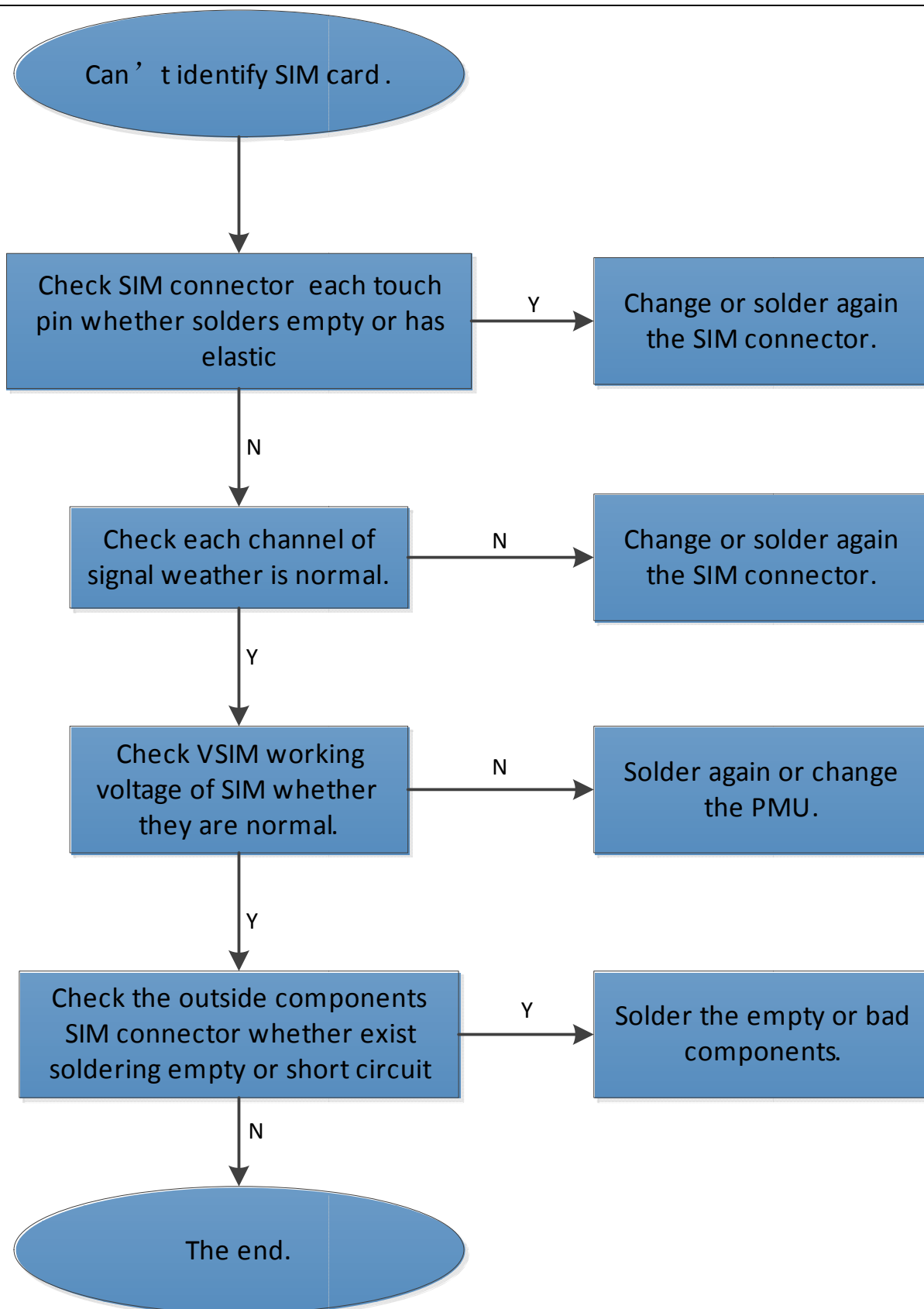




SIM1 and SIM2 power supply and signal is separated, when the SIM holder does not recognize the SIM card can check the relevant circuit. Fault, not card analysis process is as follows:

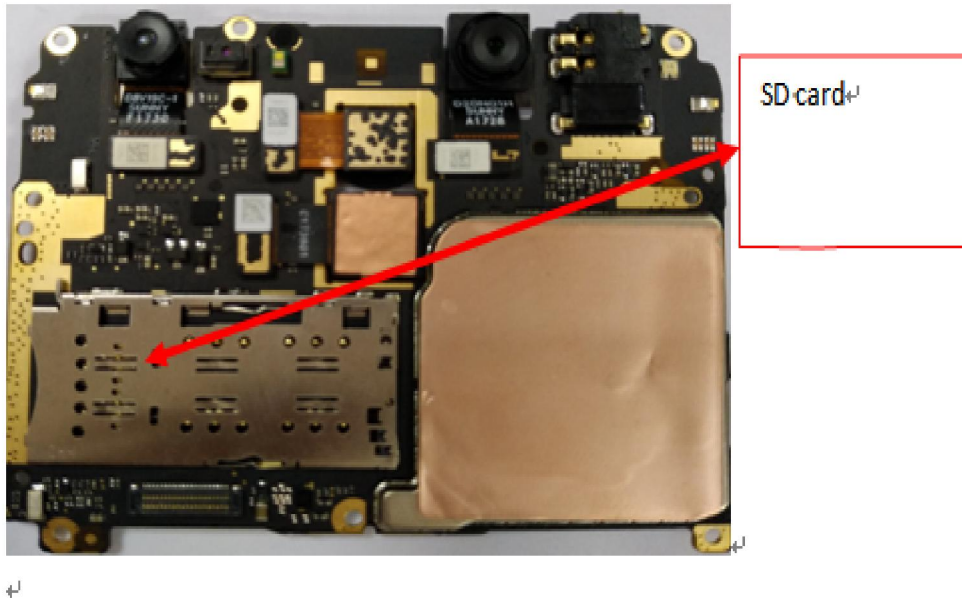
1. The first SIM booth of welding and examine whether the pin height consistent.
2. Review the working voltage of SIM card 'VDDSIM' is normal.
3. Check the SIM card control signals' SIM_SCLK ', 'SIM_DA', lines is normal.
4. No problem with the above checks, the CPU is replaced.

SIM card Issue repair flow:



14: T card issue

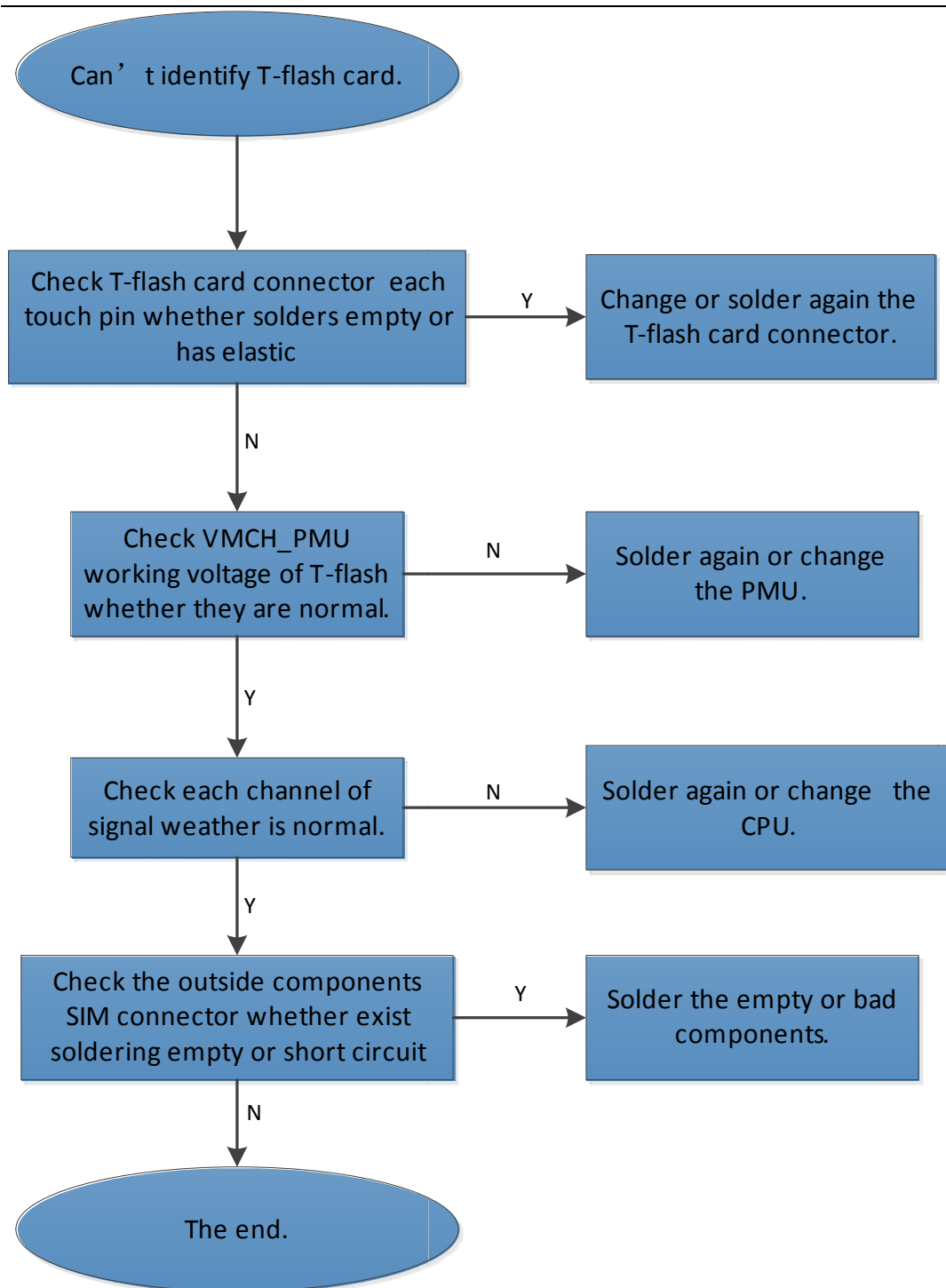
Block Diagram



T card analysis process is as follows:

1. Check if the working voltage of T card 'VDDSD0' normal and abnormal check line device on the welding is normal, then change - PMU - 2.
2. Review the T card control signals' SDO_DAT0-3 ', 'SDO_CMD' and 'SDO_CLK' on a line welding device is normal.
3. No problem with the above checks, the CPU is replaced.

T card Issue repair flow:



15. demolition TP step.

1. Prepare gun heating stations or large network, the temperature setting of 100 degrees. The display on the heating stage is heated (100 ° Winds gale execution set aligned second gear display four weeks heating) When the display is heated to about 75-80 degrees, remove the display screen using a suction device with suction of TP Pull. Let the screen frame double-sided adhesive slowly opened,
2. You can also use a small blade to pry loose TP four weeks, when the double-sided TP username viscosity becomes TP hours can be removed as follows:



TP heated aligned four weeks.

Use suction screen pair IP for suction.



